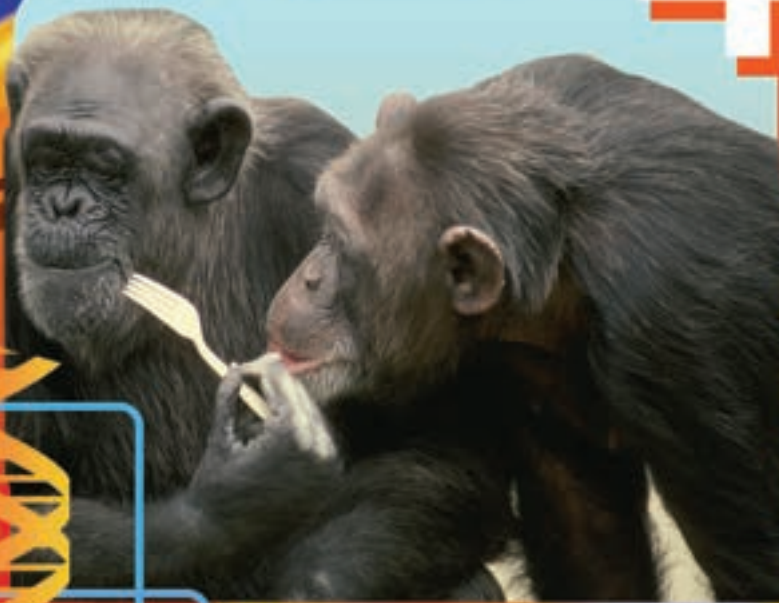


Practical Experiments in

BIOLOGY

4



Satya Naidu



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BIOLOGY

4

Satya Naidu

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Acknowledgements

I wish to thank Carol Andrews for reviewing the original drafts and Pam Robinson for her suggestions and for providing materials for the experiments.

K. Satya Naidu

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Publisher: Susan Watson
Editor: Ruth Schultz
Page design: Ruth Schultz
Page formatting and diagrams: Replika Press
Illustrations: Infographics Pty Ltd
Cover design: Gail McManus
Printed in Australia by Imprinta Pty Ltd, Hawthorn

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108 Main Road
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ISBN 978-1-92561-013-0

INTRODUCTION

Practical Experiments in Biology Book 4 is part of a four-book series that presents experiments from different areas of study that are relevant to senior Biology students. Book 4 focuses on genetics and evolution.

Teacher's notes and useful websites for each experiment are provided at the start of the book. These include strategies for enhancing student understanding of the experiment and suggestions for overcoming difficulties. Internet addresses for relevant websites are also provided to encourage students to undertake further research.

The experiments are presented in a self-contained format that Biology students will find easy to follow. Each experiment includes a summary of relevant background knowledge and is structured according to aim, hypothesis, equipment, safety rules, method, results, analysis of results and conclusion. Tables and space for diagrams are provided for students to record their results. The clear, step-by-step structure will encourage the efficient use of experimental time and give guidance to students on how to write scientific reports. Suggested answers and expected results have also been provided where appropriate.

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TEACHER'S NOTES AND USEFUL WEBSITES

Experiment 1: Extraction of DNA from onions

Suggestions for teachers

- Students should have a good background knowledge of the structure and function of DNA.
- All the materials used should be as fresh as possible.
- Students often pour ethanol too quickly. This mixes the two layers instead of forming two distinct layers. When this happens, DNA does not precipitate.
- Do not use laboratory filter paper as the liquid will not pass through it. Cheesecloth or a reusable coffee filter can be used to strain the materials.
- Fresh papaya or pineapple juice can be substituted for the meat tenderiser solution. (Use the same amount of solution as given for the meat tenderiser solution.)
- Make students aware of appropriate websites.

Websites

<http://gslc.genetics.utah.edu/units/activities/extraction/>
http://www.accessexcellence.org/AE/ACC/DNA_extractions.html#onion
http://www.exploratorium.edu/ti/human_body/dna.html
<http://www.ebioinfogen.com/dnaextraction.htm>
http://biotech.biology.arizona.edu/labs/DNA_extraction_onion_stud.html

Experiment 2: Modelling a DNA molecule and DNA replication

Suggestions for teachers

- Explain that this model is simpler than the model made by James Watson and Francis Crick. There are fewer pieces in this model than in a single strand of DNA. The pieces have simpler shapes than the true components of DNA. This model is two-dimensional, whereas the true structure of DNA is three-dimensional.
- Tell the students that there is no 'correct' sequence of DNA model bases. The activity is designed to illustrate that the DNA bases can fit into the DNA structure in any sequence.
- Make sure that the students try several possible sequences of DNA model bases before gluing their final DNA model together.
- Remind the students that one strand goes down in one direction and the second strand goes in the opposite direction.
- Extend the activity by constructing a three-dimensional model of DNA. For example, try using lego bricks or beads and pipe cleaners.
- Make students aware of appropriate websites.

Websites

http://www.accessexcellence.org/AB/GG/dna_replicating.html
<http://www.pbs.org/wgbh/aso/tryit/dna/indexdna.html>
<http://gslc.genetics.utah.edu/units/basics/builddna/>
<http://faculty.clintoncc.suny.edu/faculty/Michael.Gregory/files/Bio%20101/Bio%20101%20Lectures/biochemistry/DNA.jpg>
http://www.phy.cam.ac.uk/camphy/outreach/dna/making_models_of_dna.pdf

<http://www.ncc.gmu.edu/dna/replicat.htm>
<http://www.johnkyrk.com/DNAanatomy.html>

Experiment 3: Modelling protein synthesis

Suggestions for teachers

- Students should have good background knowledge of DNA structure, replication and protein synthesis.
- Students should complete modelling of DNA replication before this activity.
- Have the students observe a model, chart and video on protein synthesis before this activity.
- Make sure students understand the differences between DNA, mRNA and rRNA, and rRNA and tRNA. Students should be aware that in mRNA and tRNA, the nitrogenous base T is replaced by U.
- Explain to students how to use the codon chart.
- Emphasise that students must use the codons on mRNA to find the amino acids formed.
- Make students aware of appropriate websites.

Websites

<http://www.people.virginia.edu/~rjh9u/trtrpict.html>
<http://www.brooklyn.cuny.edu/bc/ahp/BioInfo/SD.TransTrans.HP.html>
<http://www.umaine.edu/fes/Classes/fes100/maps/map11diag.htm>
<http://www.ncc.gmu.edu/dna/ANIMPROT.htm>
<http://gslc.genetics.utah.edu/units/basics/transcribe/>
<http://www.tokyo-med.ac.jp/genet/mutation/nn.gif>
<http://www.biology.ewu.edu/aHerr/Genetics/Bio310/PDFs/HwPDFs/Transcribe.pdf>
<http://www.cat.cc.md.us/biotutorials/protsyn/peptidea.html>
<http://www.pbs.org/wgbh/aso/tryit/dna/indexpro.html>
http://www.contexto.info/DNA_Basics/Protein_synthesis.htm
<http://ghs.gresham.k12.or.us/science/ps/sci/ibbio/chem/nucleic/chpt15/chpt15.htm>

Experiment 4: Types of inheritance

Suggestions for teachers

- Students must have a good background knowledge of complete dominance, incomplete dominance, co-dominance and sex-linked recessive inheritance.
- Make sure students understand the terms: genotypes, phenotypes, homozygous and heterozygous conditions.
- Show videos and computer simulations about modes of inheritance to the class before the activity.
- Explain to the students how to use punnet squares to work out genotypes and phenotypes.
- Make students aware of appropriate websites.

Websites

<http://en.wikipedia.org/wiki/Snapdragon>
<http://www.blc.arizona.edu/courses/181gh/rick/genetics3/modified.html>
http://projects.cbe.ab.ca/Diefenbaker/Biology/Bio%20Website%20Final/notes/classical/3_incomplete_dominance.htm
http://natsci.edgewood.edu/Zoology%20101/beyond_mendel.htm

Experiment 5: Pedigree analysis**Suggestions for teachers**

- Make sure that students have good background knowledge about modes of inheritance and the interpretation of pedigree charts.
- Have the students practise drawing pedigree charts.
- Make students aware of appropriate websites.

Websites

http://www.blc.arizona.edu/courses/181gh/rick/human_genetics/pedigree.html
http://www.biology.arizona.edu/human_bio/problem_sets/human_genetics/01Q.html
<http://www.ndsu.nodak.edu/instruct/mcclean/plsc431/mendel/mendel9.htm>
<http://www.hhmi.org/biointeractive/vlabs/cardiology/content/dtg/pedigree/pedigree.html>
<http://www.uic.edu/nursing/genetics/Lecture/Types/Eval/typesevalans.htm>
http://www.riverdeep.net/science/biology_gateways/bg_handouts/gen/gend2tni.pdf

Experiment 6: Dihybrid inheritance**Suggestions for teachers**

- Students must have good background knowledge of Mendelian inheritance and the use of punnet squares.
- Ask students to practise monohybrid, dihybrid crosses and test crosses before doing this experiment.
- Make students aware of appropriate websites

Websites

<http://www.ndsu.nodak.edu/instruct/mcclean/plsc431/mendel/mendel3.htm>
http://anthro.palomar.edu/mendel/mendel_1.htm
http://www.longman.co.uk/gcsebiology/pdfs/worksheet_6_mendels_peas.pdf
http://www.accessexcellence.org/AE/AEC/AEF/1996/nolin_pea.html

Experiment 7: Patterns of evolution**Suggestions for teachers**

- Students should have a good understanding of the different types of evolution and the differences between homologous and analogous structures.
- Show the students a video about patterns of evolution before this experiment.
- Make students aware of appropriate websites.

Websites

<http://techcenter.davidson.k12.nc.us/spring026/anaat.htm>
<http://www.sparknotes.com/biology/evolution/patternsofevolution/section1.html>
<http://animaldiversity.ummz.umich.edu/site/accounts/information/Metatheria.html>
http://www.pbs.org/wgbh/evolution/library/01/4/pdf/i_014_02.pdf
<http://www.txtwriter.com/Backgrounders/Evolution/EVpage14.html>

Experiment 8: Natural selection in peppered moths**Suggestions for teachers**

- Students should have a good understanding of evolution,

natural selection, mutation and Hardy-Weinberg Law and its applications.

- Students must know how to use Hardy-Weinberg equations prior to this activity.
- Make students aware of appropriate websites.

Websites

http://users.rcn.com/jkimball.ma.ultranet/BiologyPages/H/Hardy_Weinberg.html
<http://staff.jccc.net/pdecell/bio205/webct/natselmodels/hardylab.html>
http://bioweb.usc.edu/courses/2001-fall/documents/blsc111-arnheim_20.pdf
<http://www.isu.edu/biolearn/Lesson%20Plans/evolution/lessonplans/LessonPopulationGenetics.html>
<http://rio.uwinnipeg.ca/~simmons/lb5pg4.htm>
http://en.wikipedia.org/wiki/Peppered_moth#Rise_and_fall_of_phenotype_frequency
http://www.am.org/docs/wells/jw_pepmoth.htm

Experiment 9: Adaptive radiation of finches and tortoises in Galapagos Islands**Suggestions for teachers**

- Students must have good background knowledge about types of evolution (convergent evolution, divergent evolution and co-evolution), genetic drift, founder effect, selective pressures, natural selection and speciation before doing this experiment.
- Show the students a video on evolution and natural selection.
- Make students aware of appropriate websites.

Websites

<http://www.rit.edu/~rhrsbl/GalapagosPages/DarwinFinch.html>
<http://www.abdn.ac.uk/zoohons/lecture5/tsld001.htm>
<http://www.discovergalapagos.com/tortoise.html>
<http://www.gct.org/tortoise.html>
<http://www.rit.edu/~rhrsbl/GalapagosPages/Tortoise2.html>
http://www.pbs.org/wgbh/evolution/library/01/6/image_pop/i_016_02.html

Experiment 10: Comparison of primate features**Suggestions for teachers**

- Students should have good background knowledge about structural changes during primate evolution.
- Provide charts, specimens and models of apes, australopithecine and homo species.
- Show the students a video on primate evolution and the discovery of Lucy.
- Make students aware of appropriate websites.

Websites

<http://www.talkorigins.org/faqs/homs/crania.html>
<http://www.kcmetro.edu/~westra/AnthroWeb/3PhysicalAnthro.html>
http://www.aronline.net.au/human_evolution/skeleton/more_info.htm#4
<http://www.unc.edu/courses/pre2000fall/anth100/hominids.htm>
<http://en.wikipedia.org/wiki/Human>

EXTRACTION OF DNA FROM ONIONS



Timing: 2 × 45 minute lessons

Background

The cells of all organisms contain **deoxyribonucleic acid (DNA)** in the **nucleus**. DNA is the chemical that carries **genetic code** and controls **protein synthesis**. It is made up of sub-units called **nucleotides**. Nucleotides are made of a deoxyribose sugar molecule, a phosphate group and one of four possible nitrogen bases.

There are three basic steps in **DNA extraction**. The cell must be **lysed** (broken open) to release the nucleus. Then the nucleus must be opened to release the DNA. At this point, the DNA must be protected from enzymes that will degrade it and cause shearing. Once the DNA is released, it is **precipitated** in alcohol.

In order for the cell to be lysed, the lipid walls must be broken down. The detergent and salt solutions accomplish this. The cell walls, cell membranes, and nuclear membranes are broken down by the action of the blender. The meat tenderiser contains **papain**, an enzyme that helps remove the protein that can contaminate the DNA. The **ethanol** is used to precipitate the DNA. DNA is soluble in water. When it is placed in ethanol, DNA uncoils and precipitates, leaving behind the other cell components that are not soluble in ethanol.

Aim To extract DNA from onion cells.

Equipment

- 3 fresh onions (medium-sized)
- knife
- graduated cylinders (10 mL and 100 mL)
- 2 test tubes
- blender
- test tube rack
- ice bucket or a beaker with ice
- strainer
- glass rod scored on one end
- filter paper or cheese cloth
- funnel
- 15 g non-iodised salt
- 5 g fresh meat tenderiser
- Phenol red indicator
- Palmolive detergent
- 250 mL beaker
- distilled water
- ice-cold 95% ethanol
- sealed tube with a lid

SAFETY RULES

- ⚠ Wear goggles and a lab coat while doing this experiment.
- ⚠ Be careful when using the blender. Make sure the lid is tightly closed before use.
- ⚠ When you have finished the experiment, dispose of materials in specified containers.

Advance preparation of solutions

Detergent/salt solution: Add 10 mL detergent and 10 g salt to 90 mL distilled water.

Enzyme solution: Add 5 g meat tenderiser to 95 mL distilled water.

5% sodium chloride solution: Add 5 g non-iodised salt to 100 mL distilled water.

Phenol red indicator: Dissolve a toothpick-end full of powdered phenol red in 100 mL distilled water. It should turn light amber in colour.

Ethanol: Place the ethanol in a plastic bottle in the freezer for at least 24 hours before the experiment. It must be stored in a sealed, vapour-tight container.

Method

- 1 Remove the skin from the 3 onions. Chop them into small pieces.
- 2 Place the onion pieces in the blender with 100 mL detergent/salt solution.
- 3 Blend on high for 1 minute. This will lyse (break up) the cell membrane and nuclear membrane.
- 4 Use filter paper placed inside a funnel to strain the mixture into a beaker.
- 5 Add 25 mL meat tenderiser to the beaker and stir to mix. This will denature any protein that could contaminate the DNA.
- 6 Place 6 mL filtrate in a test tube.

- 7 Carefully and slowly pour 10 mL ice-cold ethanol down the side of the test tube to form a layer on top of the onion mixture.
- 8 Let the mixture stand undisturbed for 2–3 minutes, until the bubbling stops.

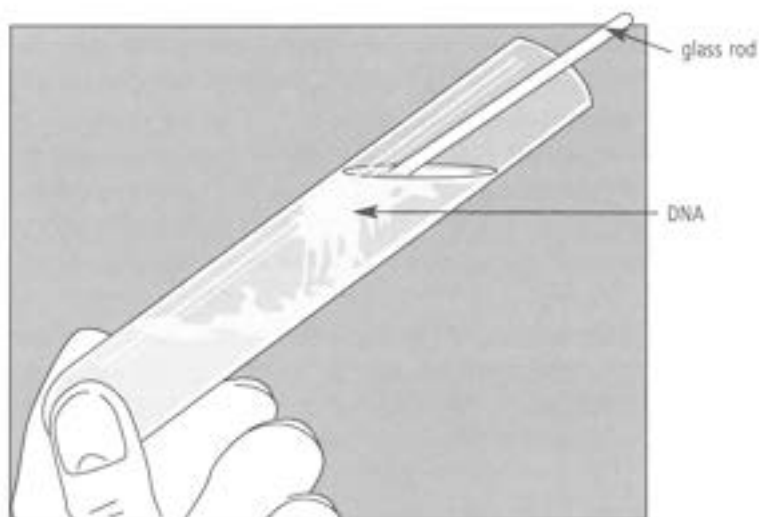


Figure 1: Extraction of DNA

- 9 DNA strands will immediately start to form and float in the alcohol. Using a twirling motion of the glass rod, slowly move the scored end of the glass rod through the interface of the two layers (the area where the alcohol and the onion mixture meet).
- 10 Collect the mucus-like DNA and place it in a test tube containing 5% sodium chloride solution. Add 5 drops of phenol red indicator to the test tube. The presence of nucleic acids in the DNA solution should turn the solution **dark pink**.
- 11 To keep the DNA, store it in 50%–70% alcohol in a sealed tube or air dry it on paper towels or filter paper.

Results

Draw a diagram of DNA after staining.

Analysis of results

- 1 Why was it necessary to blend the onion and the detergent/salt solution?
- 2 What steps were taken to eliminate protein contamination?
- 3 What happened when the ice-cold ethanol was poured into the solution?
- 4 Describe the DNA you obtained from the onion cells.
- 5 Why did the phenol red indicator turn pink?

Conclusion

- 1 Do cells of all organisms contain DNA?
- 2 Where is DNA located?
- 3 What is the function of DNA in an organism's body?

MODELLING A DNA MOLECULE AND DNA REPLICATION



Timing: 2 × 45 minute lessons

Background

Chromosomes are thread-like structures found in the **nucleus** of a cell. They are made up of **deoxyribonucleic acid (DNA)** and **proteins**. **Genes** are sections of chromosomes that code for proteins and the inherited instructions that control structure and function. The deoxyribose sugar molecule of the DNA backbone has 5 carbon atoms and 3 oxygen atoms. The carbon atoms are numbered 1', 2', 3', 4' and 5'. The hydroxyl groups on the 5'- and 3'-carbons link to the phosphate groups to form the DNA backbone.

DNA is made up of sub-units called **nucleotides**. A nucleotide consists of one nitrogen base, one phosphate group and one deoxyribose sugar molecule. There are four different nitrogen bases: **adenine**, **cytosine**, **guanine** and **thymine**. Adenine and guanine are **purines**. Cytosine and thymine are **pyrimidines**. The number of purine bases always equals the number of pyrimidine bases. Adenine always pairs with thymine and cytosine always pairs with guanine. The nucleotides make up two chains that are linked and twisted around each other in the form of a double helix.

DNA replication begins with a partial unwinding of the double helix at an area known as the **replication fork**. This unwinding is caused by the enzyme **DNA helicase**. As the two DNA strands separate, the bases are exposed. Then the enzyme **DNA polymerase** moves into position at the point where synthesis will begin. Nucleotides are added, one by one, in an exactly complementary manner: adenine to thymine and guanine to cytosine.

DNA polymerase reads the sequence of bases on the **template strand** (sense strand). Then it synthesises the **complementary strand**. The template strand is always read in the 3' to 5' direction. The new DNA strand is synthesised in the 5' to 3' direction. The two strands are described as being **antiparallel**. When the enzyme has reached the end of the DNA, there are two identical molecules of DNA. Each molecule contains one side of the original DNA and one side made of 'new' nucleotides. This is known as the **semi-conservative model**. The two new double-stranded chains then twist into two separate double helices. Thus two identical DNA molecules are formed.

Aim To make paper models of a DNA molecule and DNA replication.

Equipment

- 10 adenine, 10 thymine, 10 cytosine and 10 guanine nucleotides (each nucleotide in a different colour)
- glue stick
- adhesive tape
- 2 sheets of white, A4 paper
- scissors
- video and chart about DNA replication

SAFETY RULES

- ⚠ Be careful when cutting the cardboard pieces.
- ⚠ Make sure the shapes are accurate and fit properly.

The DNA molecule has two strands: the **sense strand** and the **antisense strand**. These strands are exact mirror images of each other, but only the sense strand contains the information for making proteins. The antisense strand does not. Antisense DNA is also known as **non-coding DNA**.

Part A – Modelling a DNA molecule

Method

- 1 Take 5 of each type of nucleotide, one sheet of white A4 paper, the glue stick, the tape and the scissors. Label each nitrogen base with its initial letter: A, T, G or C.
- 2 Examine the way the nitrogen bases are bonded together (see Figure 1). Arrange the nucleotides into a ten-rung model segment of DNA. The **sense strand** on the left must read: A T C G A C C G T G. The **antisense strand** on the right must have the complementary nitrogen bases. **Note:** Remember that adenine bonds to thymine and guanine bonds to cytosine.
- 3 Attach each pair of nitrogen bases with short pieces of tape.



- Use tape to attach the deoxyribose molecules and the phosphate groups to each strand. Make sure you attach the pieces at the correct angles (as shown in Figure 1) or your DNA model will not fit together properly.

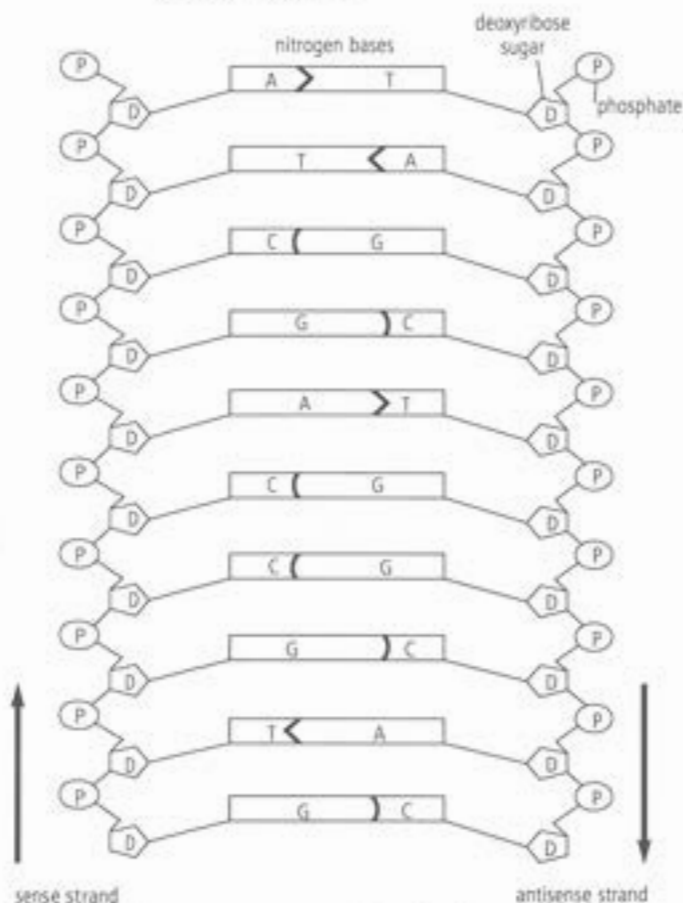


Figure 1: DNA molecule after untwisting

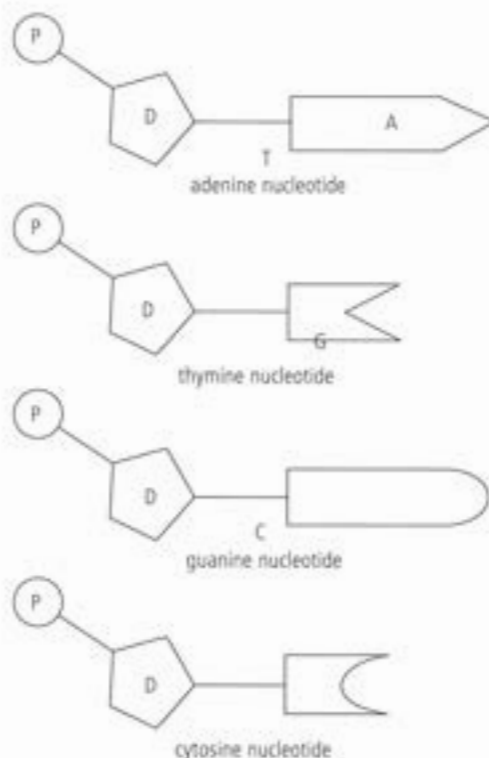


Figure 2: Nucleotides

Results

Submit your DNA model to your teacher.

Analysis of results

- What is the relationship between chromosomes and genes?
- What is the relationship between genes and DNA molecules?
- What determines the sequence of the nitrogen bases in a new DNA strand?
- Write out the sequence of the new DNA segment that would form next to the segment containing G G A C T G T T A.

Part B - Modelling DNA Replication

Method

- Lay your DNA model flat on the table. Starting at one end of the model, cut the tape that connects each pair of nitrogen bases on all the rungs.

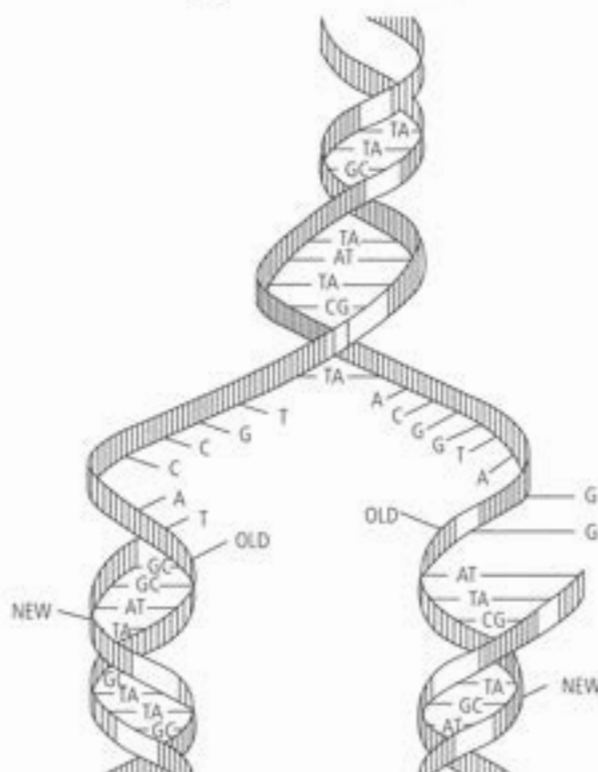


Figure 3: DNA replication

- 2 Lay the separated model aside. Take 5 of each type of nucleotide. Label each nitrogen base with its initial letter: A, T, G or C.
- 3 Matching adenine with thymine and guanine with cytosine, arrange new nucleotides on both of the two strands of your DNA model. Fasten them in place with tape. Make sure you attach the pieces at the correct angles, as shown in Figure 1.

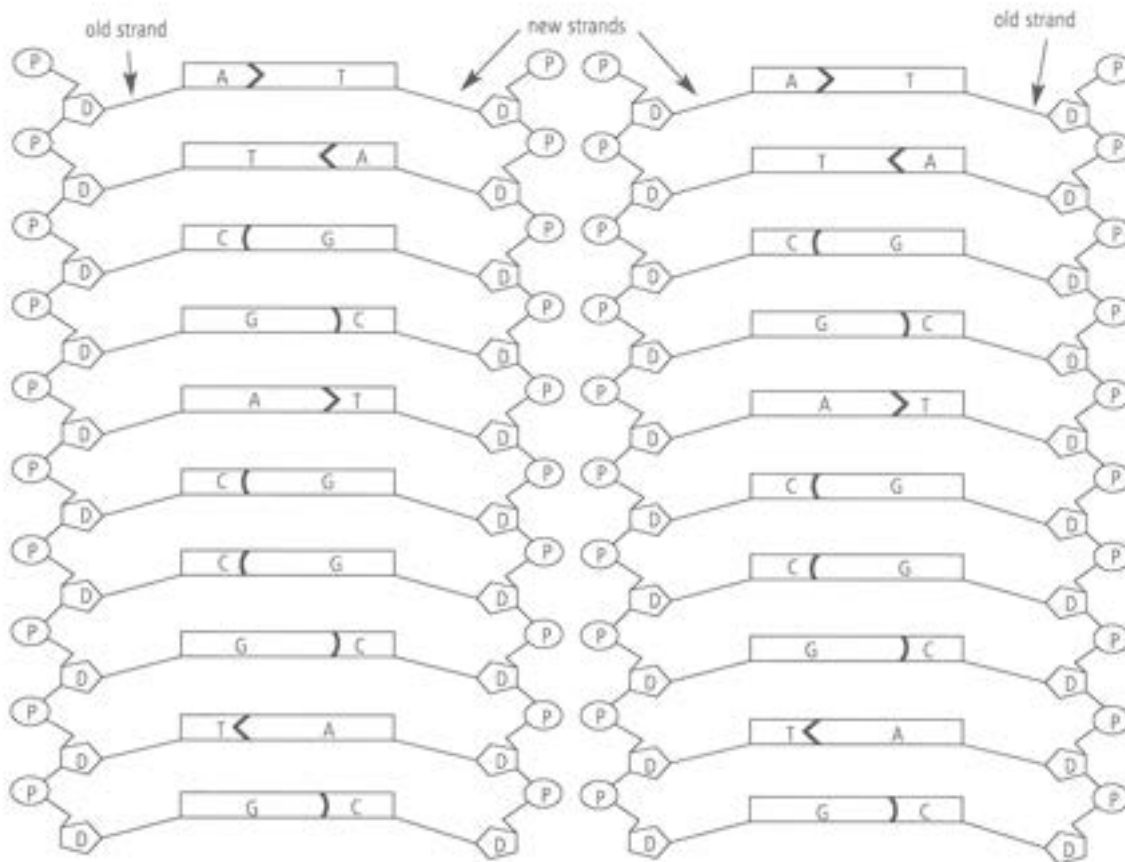


Figure 4: DNA after replication

Results

Submit your model of DNA replication.

Analysis of results

- 1 Why is this model of DNA replication known as semi-conservative?
- 2 You have made a two-dimensional model of the structure of DNA. The true three-dimensional structure of DNA is a double helix, which looks like a twisted ladder. How could you make a three-dimensional model of the structure of DNA?

Conclusion

- 1 Briefly explain the structure of DNA in your own words.
- 2 How does DNA replicate during cell division?

MODELLING PROTEIN SYNTHESIS



Timing: 2 × 45 minute lessons

Background

Protein synthesis involves two processes: **transcription** and **translation**. During transcription, a small section of the DNA double helix unwinds and the bases of the two strands are exposed. The DNA strand that is transcribed is called the **template strand** or **sense strand**. The DNA strand that is not transcribed is called the non-sense strand or **antisense strand**.

During transcription, RNA polymerase enzyme binds to the DNA molecule at a region known as the **promotor site**. This site is in front of the gene where transcription will begin. A gene consists of coding regions called **exons** and non-coding regions called **introns**. **RNA nucleotides** (ribonucleotides) line up in the proper order by hydrogen-bonding to their complementary bases on DNA. DNA that includes all the exons and introns of the gene is first transcribed into a complementary RNA copy called **nuclear RNA**. Then introns are removed from the nRNA by a process called **RNA splicing**. This edited sequence is called **messenger RNA** or **mRNA**. mRNA moves from the nucleus to the cytoplasm and attaches to a ribosome. Three-dimensional, inverted cloverleaf-shaped molecules called **tRNAs** are present in the cytoplasm. At the top end of tRNA, a specific amino acid can be attached with enzymes called **aminoacyl-tRNA synthetases**. At the lower end, the tRNAs carry **anticodons** which are complementary to mRNA **codons**.

Translation occurs at **ribosomes** in the **cytoplasm**. As protein synthesis begins, one small sub-unit and one large sub-unit of the ribosome come together to form an active ribosome. The first **A-U-G** codon on the 5' end of the mRNA acts as a '**start**' signal for the translation and codes for the introduction of a **methionine amino acid**. Amino acids continue to be linked until the protein is finished. A special type of bond is formed between amino acids by the enzyme **peptidase**. The elongation procedure continues until the proper protein is completed. A '**stop**' codon (**U-A-A**, **U-G-A** or **U-A-G**) signals the end of the process. An enzyme called the **releasing factor** then frees the newly made protein from the last tRNA. All mRNA messages are eventually degraded when there is no need to make the protein anymore.

Aim To model the transcription and translation of DNA during protein synthesis

Equipment

- white paper
- coloured pens
- chart and video on protein synthesis

Method

- 1 Draw a large, green circle on a sheet of paper. Label the circle 'cell membrane'. Use purple to draw a smaller circle inside it. Label this circle 'nuclear membrane'.
- 2 Use blue to draw the template strand of the DNA consisting of the bases ACG TTC CCG GCC TCG.
- 3 Use red to construct the mRNA with the complementary bases. **Note:** in mRNA, T (thymine) is replaced by U (uracil).
- 4 Draw the mRNA coming out of the nuclear membrane and attached to the ribosome. (See Figure 4.) Use pink to draw the ribosome.
- 5 Use black to draw a tRNA. (See Figure 2.) Write the anticodon relevant to the first codon.

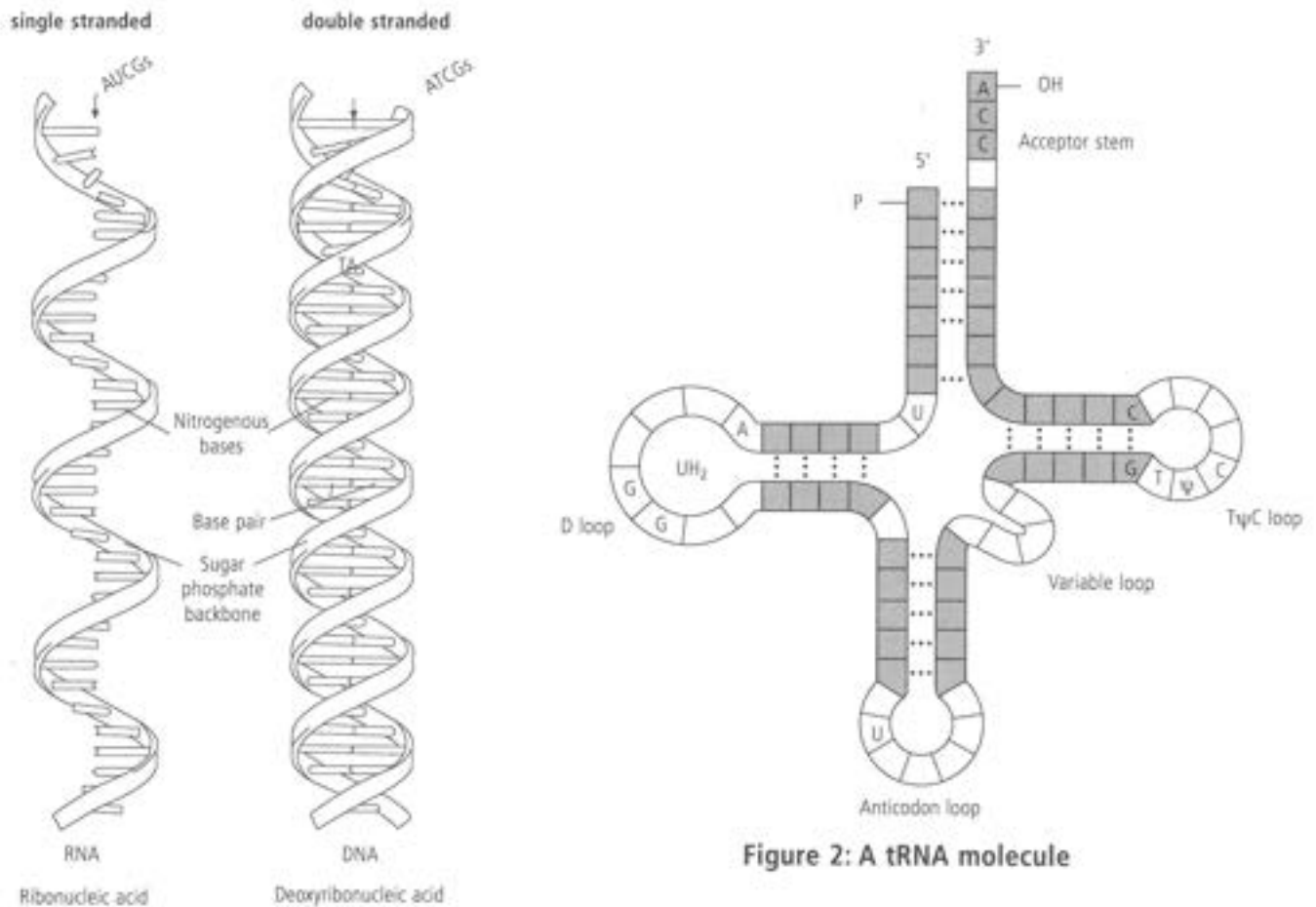


Figure 1: DNA and RNA strands

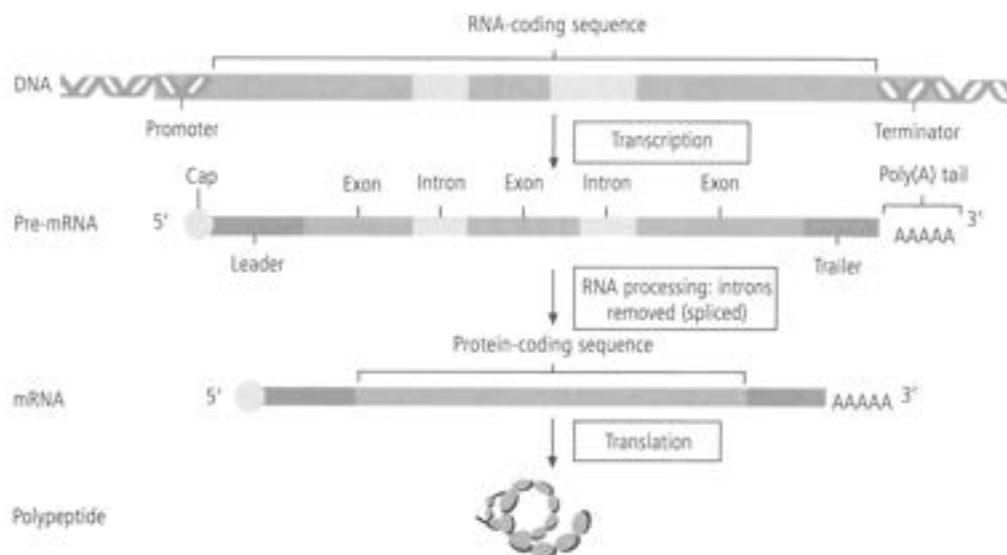


Figure 3: Transcription and translation

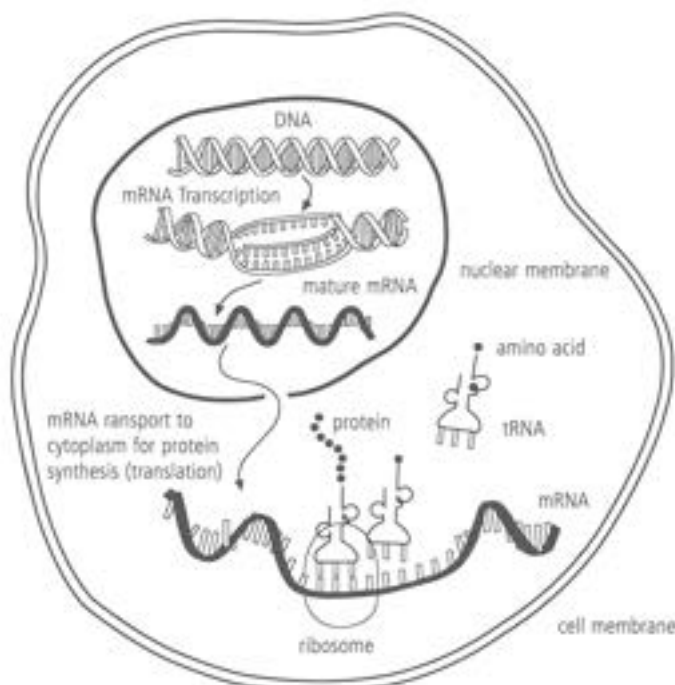


Figure 4: Protein synthesis

Second base					
First base	U	C	A	G	Third base
	UUU Phenylalanine (Phe)	UCU Serine (Ser)	UAU Tyrosine (Tyr)	UGU Cysteine (Cys)	
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	
	UUA Leucine (Leu)	UCA Ser	UAA STOP	UGA STOP	
C	UUG Leu	UCG Ser	UAG STOP	UGG Tryptophan (Trp)	G
	CUU Leucine (Leu)	CCU Proline (Pro)	CAU Histidine (His)	CGU Arginine (Arg)	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Glutamine (Gln)	CGA Arg	A
A	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
	AUU Isoleucine (Ile)	ACU Threonine (Thr)	AAU Asparagine (Asn)	AGU Serine (Ser)	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lysine (Lys)	AGA Arginine (Arg)	A
G	AUG Methionine (Met) or START	ACG Thr	AAG Lys	AGG Arg	G
	GUU Valine Val	GCU Alanine (Ala)	GAU Aspartic acid (Asp)	GGU Glycine (Gly)	U
	GUC (Val)	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glutamic acid (Glu)	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

Figure 5: Genetic code chart (mRNA codons)

DNA template strand	mRNA sequence	tRNA anticodons	Amino acids
A			
C			
G			
T			
T			
C			
C			
C			
G			
G			
C			
G			
T			
G			
C			

Results

- 1 Submit your model of protein synthesis.
- 2 Use the codon chart (Figure 5) to complete the amino acids formed in the table above.

Analysis of results

- 1 Where in the cell does transcription take place?
- 2 How is a mRNA strand different from a DNA strand?
- 3 Which strand acts as the template for mRNA transcription?
- 4 What are the functions of rRNA, tRNA and mRNA?
- 5 What is a codon? What is an anticodon? Give one example of each.
- 6 How is the sequence of amino acids in the protein determined?
- 7 Use the codon chart to complete the following tables.

Table 1

Template strand				GGG	
Non-template strand	UAC				
mRNA					CCU
tRNA		UCG			
Amino acid			Leu		

Table 2

mRNA codon	tRNA anticodon	Amino acid	mRNA codon	tRNA anticodon	Amino acid
AUU			UGG		
GGA			GAA		
AUG			GAU		
UCU			CCC		

Conclusion

In your own words, summarise how proteins are synthesised.

TYPES OF INHERITANCE



Timing: 2 × 45 minute lessons

Background

In genetics, **dominance relationships** are the ways genes interact to express themselves as phenotypes in an individual. There are three kinds of dominance relationships. **Simple dominance** is where one gene's allele is completely dominant over the recessive allele. The individual is thus a carrier of the recessive gene but the recessive gene is not expressed at all. **Incomplete dominance** is where both alleles of a heterozygote influence the phenotype. A cross between organisms with two different phenotypes produces offspring with a third phenotype that is a blending of the parental traits. For example, when red snapdragon plants are crossed with white snapdragon plants, they produce pink flowers.

Codominance is the situation in which a heterozygote shows the phenotypic effects of both alleles equally. A cross between organisms with two different phenotypes produces offspring with a third phenotype in which both of the parental traits appear together. Among the alleles controlling human ABO blood groups, the alleles for A group and B group are codominant and are expressed in the offspring with AB blood group. The allele for O group is recessive to both A and B alleles.

There are two types of **sex linked inheritance**: Sex-linked recessive inheritance and sex linked dominant inheritance. In **sex-linked recessive inheritance**, the recessive gene is located on the X chromosome. If the mother is a carrier, her sons will be affected by the trait and the daughters could become carriers. Colour blindness, haemophilia and Duchene muscular dystrophy are examples of sex-linked recessive disorders. In **sex-linked dominant inheritance** if the father has the trait then all his daughters will be affected. However sons are not affected because sons inherit Y from the father and X from the mother. If the mother is affected, 50% of her sons and daughters will be affected. In **sex-linked dominant inheritance**, affected males transmit the trait to all their daughters and none of their sons. If a mother has an X-linked dominant disorder, there is a 50% chance that each of her children will be affected.

Part A – Incomplete dominance in snapdragon plants

Aim To work out the genotypes and phenotypes of offspring produced in F₁ and F₂ generations when a red snapdragon is crossed with a white snapdragon.

Equipment • punnet squares • diagrams of snapdragon flowers • pen and paper

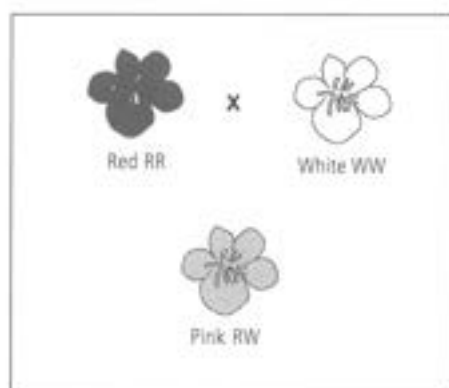


Figure 1: Red snapdragon crossed with white snapdragon

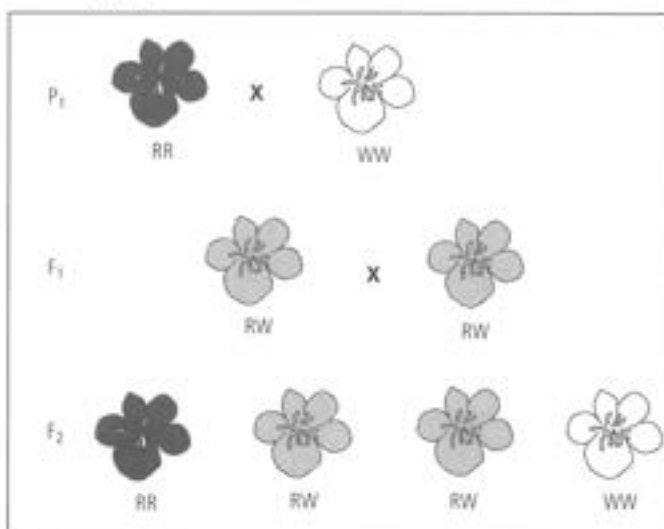


Figure 2: Pink snapdragon crossed with pink snapdragon

Method

Use punnet squares to work out the genotypes and phenotypes produced when the following crosses are carried out. Record your work under Results.

- 1 Red snapdragon (RR) crossed with white snapdragon (WW)
- 2 Pink snapdragon (RW) crossed with pink snapdragon (RW)
- 3 Red snapdragon (RR) crossed with pink snapdragon (RW)

Results**Table 1** Red × white

W h i t e	Red		
		R	R
	W		
	W		

Genotypes of F1: _____

Phenotypes of F1: _____

Table 2 Pink × pink

P i n k	Pink		
		R	W
	R		
	W		

Genotypes of F1: _____

Phenotypes of F1: _____

Table 3 Red × pink

R e d	Pink		
		R	W
	R		
	R		

Genotypes of F1: _____

Phenotypes of F1: _____

Analysis of results

- 1 Which two colours are incompletely dominant in snapdragons?
- 2 Refer to your results to explain what happened when
 - a a red snapdragon was crossed with a white snapdragon
 - b two pink snapdragons were crossed
 - c a red snapdragon was crossed with a pink snapdragon.

Part B – Co-dominance in ABO blood groups**Aim**

To work out the genotype and phenotype ratios of the offspring produced when different blood groups are crossed.

Blood transfusions

People with the AB blood group can accept any type of blood: A, B, AB, or O. People with the O blood group can only accept blood from another person with O type blood. However, they can give blood to all the other groups. The AB blood group is known as the **universal receiver**. The O blood group is known as the **universal donor**.

Method

Use punnet squares to work out the genotypes and phenotypes produced when the following matings occur between people with the specified blood groups. Record your work under Results.

- 1 A homozygous A group ($I^A I^A$) with a homozygous A group ($I^A I^A$)
- 2 A heterozygous A group ($I^A I^O$) with a heterozygous A group ($I^A I^O$)
- 3 A homozygous A group ($I^A I^A$) with a homozygous B group ($I^B I^B$)
- 4 A heterozygous A group ($I^A I^O$) with a homozygous O group ($I^O I^O$)
- 5 A heterozygous B group ($I^B I^O$) with a homozygous O group ($I^O I^O$)
- 6 A homozygous O group ($I^O I^O$) with a homozygous O group ($I^O I^O$)

Results**Table 4** Homozygous group A × homozygous group A

A g r o u p	A group		
		I^A	I^A
	I^A		
	I^A		

Genotypes: _____

Phenotypes: _____

Table 5 Heterozygous group A × heterozygous group A

A g r o u p	A group		
		I^A	I^O
	I^A		
	I^O		

Genotypes: _____

Phenotypes: _____

Table 6 Homozygous group A × homozygous group A

B g r o u p	A group		
		I^A	I^A
	I^B		
	I^B		

Genotypes: _____

Phenotypes: _____

Table 7 Heterozygous group A × homozygous group O

O g r o u p	A group	
	I^A	I^A
	I^O	
	I^O	

Genotypes: _____

Phenotypes: _____

Table 8 – Heterozygous group B × homozygous group O

O g r o u p	B group	
	I^B	I^O
	I^O	
	I^O	

Genotypes: _____

Phenotypes: _____

Table 9 Homozygous group O × homozygous group O

O g r o u p	O group	
	I^O	I^O
	I^O	
	I^O	

Genotypes: _____

Phenotypes: _____

Analysis of results

- 1 Which two alleles are co-dominant?
- 2 Which alleles are dominant to the O allele?
- 3 Which genotypes and phenotypes are produced when the following matings occur?
 - a 2 homozygous group As are crossed
 - b 2 heterozygous group As are crossed
 - c A homozygous group A is crossed with a homozygous O group
 - d A heterozygous group A is crossed with a homozygous O group
 - e A heterozygous group A is crossed with a heterozygous B group
 - f 2 group Os are crossed

Part C – Sex-linked recessive and dominant inheritance

Aim To investigate the type of inheritance for colour-blindness, haemophilia and brown tooth enamel.

Method

Use punnet squares to work out the genotypes and phenotypes produced when the following matings occur between people affected with colour-blindness, hemophilia and brown teeth enamel. Record your work under Results. **Note:** Underline the small ς to distinguish it from the capital C.

- 1 **Colour-blindness: sex-linked recessive**
 - a Colour-blind male × normal female ($X^cY \times X^CX^C$)
 - b Normal male × carrier female ($X^CY \times X^CX^c$)
- 2 **Haemophilia: sex-linked recessive**
 - a Haemophiliac male × carrier female ($X^hY \times X^HX^h$)
 - b Normal male × carrier female ($X^HY \times X^HX^h$)
- 3 **Brown teeth enamel: sex-linked dominant**
 - a Male with brown teeth enamel × female with white teeth enamel ($X^EY \times X^eX^e$)
 - b Male with white teeth enamel × heterozygous female with brown teeth enamel ($X^eY \times X^EX^e$)

Results

Table 10 Colour-blindness: Colour-blind male × normal female

	X^cY	
	X^c	Y
	X^CX^c	
	X^c	
	X^c	

Genotypes: _____

Phenotypes: _____

Table 11 – Colour-blindness: Normal male × carrier female

	X^CY	
	X^C	Y
	X^CX^c	
	X^c	
	X^c	

Genotypes: _____

Phenotypes: _____

Table 12 Hemophilia: Hemophiliac male × carrier female

	X^hY	
	X^h	Y
	X^HX^h	
	X^H	
	X^h	

Genotypes: _____

Phenotypes: _____

Table 13 Hemophilia:
normal male $\times$ carrier female

X^{Hh}	$X^{Hh}Y$		
	X^H	Y	
	X^h		

Genotypes: _____

Phenotypes: _____

Table 14 Brown teeth enamel: male with brown teeth enamel $\times$ female with white enamel

$X^E X^e$	$X^E Y$		
	X^E	Y	
	X^e		

Genotypes: _____

Phenotypes: _____

Table 15 Brown teeth enamel: male with white teeth enamel $\times$ heterozygous female with brown teeth enamel

$X^E X^e$	$X^e Y$		
	X^e	Y	
	X^E		

Genotypes: _____

Phenotypes: _____

Analysis of results

- 1 Which alleles are dominant and which alleles are recessive for the following conditions?
 - a Colour-blindness
 - b Hemophilia
 - c Brown tooth enamel
- 2 Complete the table with the phenotypes produced in each case.

Phenotype of parents	Sex-linked recessive inheritance: colour-blindness	Sex-linked dominant inheritance: brown teeth enamel
Affected male $\times$ normal female	$X^c Y \times X^C X^C$	$X^E Y \times X^e X^e$
Normal male $\times$ carrier female	$X^C Y \times X^C X^c$	$X^e Y \times X^E X^e$
Affected male $\times$ carrier female	$X^c Y \times X^C X^c$	$X^E Y \times X^E X^e$

- 3 Why are more males affected in sex-linked recessive disorders?
- 4 Why are all the girls affected in sex-linked dominant disorders if the father has the disorder?

Conclusion

- 1 Complete the table.

	Incomplete dominance	Co-dominance	Sex-linked recessive inheritance	Sex-linked dominant inheritance
Location of the gene				
Phenotype of heterozygote				
Examples				

- 2 Which blood groups are compatible? Which group is known as universal donor? Which group is known as universal?

ANALYSIS OF HUMAN PEDIGREES



Timing: 2 × 45 minute lessons

Background

The **pattern of inheritance** for many traits is worked out using pedigree analysis. A **pedigree** is a diagram of a family that lists the members of the family and their relationships. The primary patterns of inheritance are autosomal recessive, autosomal dominant, sex-linked recessive and sex-linked dominant.

In **autosomal recessive inheritance**, the recessive gene is located on one of the **autosomes**. Unaffected parents (heterozygotes) can produce affected offspring. The affected individual must be homozygous recessive. This trait can be inherited by both males and females, and it can skip generations. Examples of autosomal recessive inheritance are cystic fibrosis, thalassemia, Tay-Sachs disease and sickle cell anaemia. In **autosomal dominant inheritance**, the dominant gene is located on one of the autosomes. Affected individuals must carry at least one dominant gene, which can then be passed on to both males and females. The inheritance of this trait does not skip generations. Examples of autosomal dominant inheritance are Huntington disorder and Marfan syndrome.

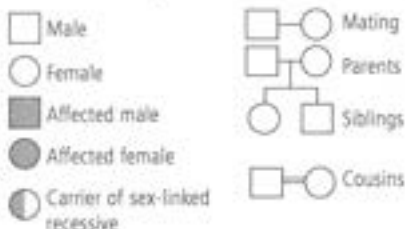
In **sex-linked recessive inheritance**, the gene is located on the **X chromosome**. More males are affected by this type of inheritance than females, because males inherit the X chromosome from their mothers and the Y chromosome from their fathers. Females can only inherit the trait if the father has it and the mother is a carrier. If females are affected or carriers, they pass the gene to their sons. Males cannot be carriers. The inheritance can skip generations. Examples of sex-linked recessive inheritance are colour-blindness, Duchene muscular dystrophy and hemophilia.

In **sex-linked dominant inheritance**, the dominant gene is located on the X chromosome. Affected males transmit the trait to all their daughters and none of their sons. If the mother has an X-linked dominant trait, there is a 50% chance that each of her children will be affected. Examples of **sex-linked dominant inheritance** are Achondroplasia (dwarfism), retinoblastoma, rickets and brown teeth enamel.

Aim To analyse pedigree charts and determine the mode of inheritance

Equipment • pedigree charts • pen and paper

Symbols used in pedigree charts

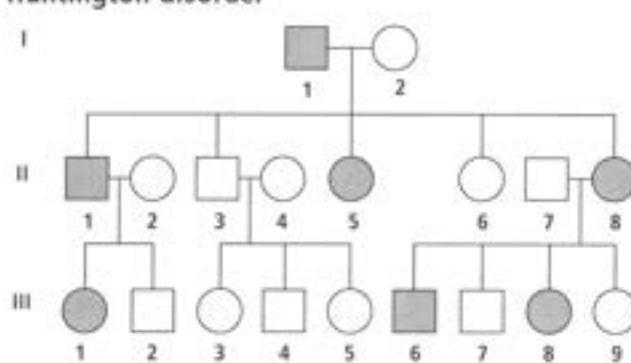
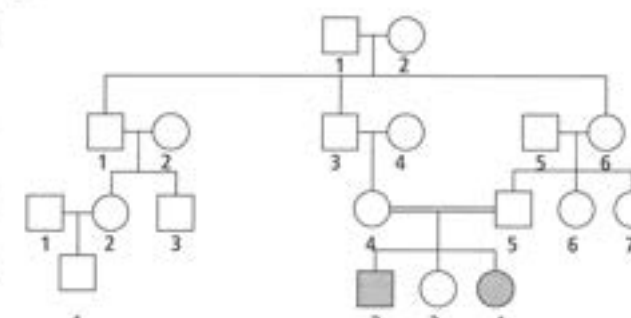
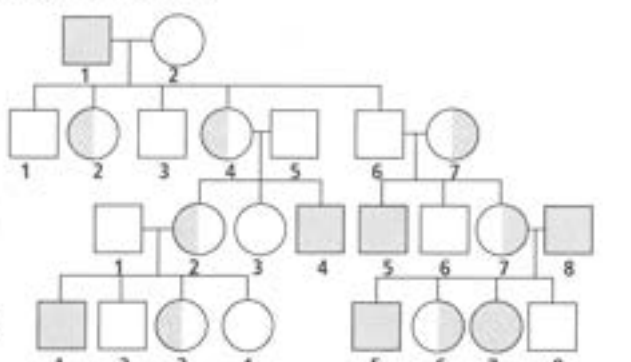
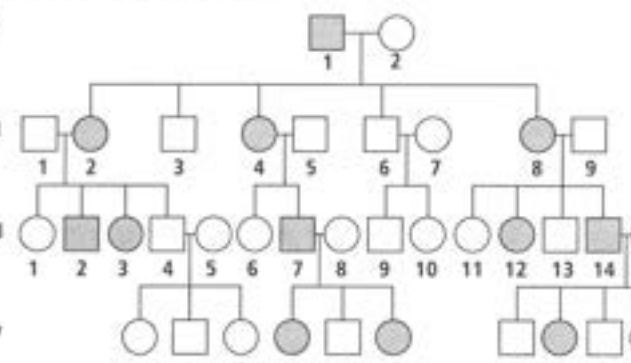


The Roman numerals (I, II, III ...) symbolise generations. The Arabic numerals (1, 2, 3 ...) symbolise the birth order within each generation. Double lines between individuals indicate that they are cousins.

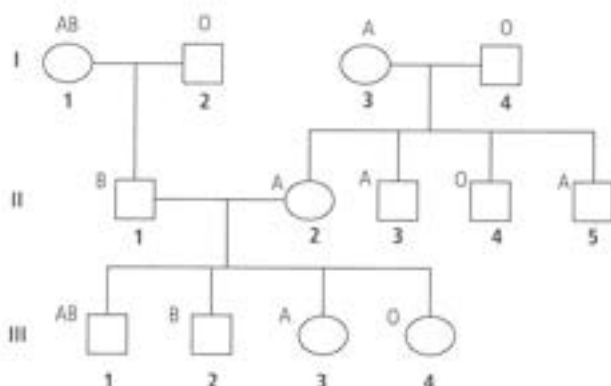
Part A – Analysis of pedigrees

Method

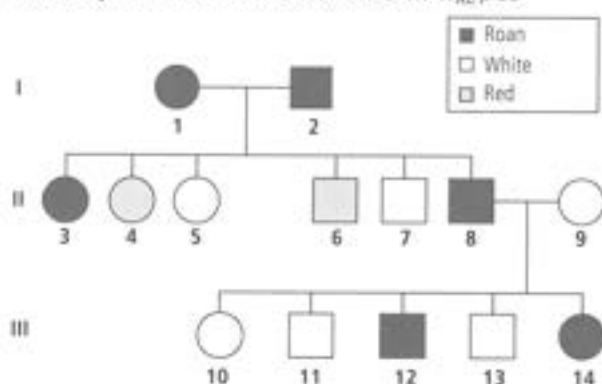
Analyse the following pedigrees and determine the mode of inheritance: autosomal dominant, autosomal recessive, sex-linked recessive, sex-linked dominant, co-dominant or incompletely dominant. Give evidence to support your answers.

Pedigree charts	Type of inheritance and analysis of pedigree chart
1 Huntington disorder 	
2 Cystic fibrosis 	
3 Colour-blindness 	
4 Brown tooth enamel 	

5 ABO blood group



6 Roan, red and white colours in horses



Analysis of results

- 1 Pedigree chart 1 shows the inheritance of Huntington disorder (HD). Use H and h symbols for alleles.
 - a What are the genotypes of parents in the first generation?
 - b How many males and females inherited Huntington disorder in the second and third generations?
- 2 Pedigree chart 2 shows the inheritance of cystic fibrosis. Use C and c symbols for alleles.
 - a What are the genotypes of the parents in the first generation?
 - b How many males and females inherited cystic fibrosis in the second, third and fourth generations?
 - c What is the relationship between III4 and III5?
- 3 Pedigree chart 3 shows the inheritance of colour blindness. Use $X^C X^C$ for colour-blind females, $X^C X^c$ for carrier females, $X^C X^C$ for normal females, $X^c Y$ for colour-blind males and $X^C Y$ for normal males.
 - a What are the genotypes of parents in the first generation?
 - b Why is a male carrying the allele for colour-blindness known as hemizygous?
 - c Why do more males inherit colour-blindness?
 - d What would be the genotypes of parents for a colour-blind girl to be produced?
 - e How many males and females are affected in the second, third and fourth generations?

- 4 Pedigree chart 4 shows the inheritance of brown tooth enamel. Use $X^E X^E$ for females with brown teeth, $X^e X^e$ for females with white teeth, $X^E Y$ for males with brown teeth and $X^e Y$ for males with white teeth.
 - a What are the genotypes of the father and mother in the first generation?
 - b Why did all the females in the second generation inherit the trait?
 - c Why did 50% of male and female offspring inherit brown teeth, when their mother had brown teeth and their father had white teeth?
 - d How many males and females are affected in the second, third and fourth generations?
- 5 In pedigree chart 5, second generation parents with A ($I^A I^O$) and B ($I^B I^O$) blood groups produced A, B, AB and O blood groups. How could this happen?
- 6 In pedigree chart 6, both parents are roan, but three types of offspring are produced. How could this happen?

Part B – Making your own family pedigree

Method

- 1 Draw an outline of your family. Show your grandparents in the first row, your parents, uncles and aunts in the second row, and then yourself, siblings and cousins in the third row.
- 2 Try to roll your own tongue. The ability to roll your tongue is a noticeable dominant trait. People who can roll their tongue have the dominant allele R. People who cannot roll their tongue have the genotype rr. Enter your genotype on the pedigree as R or rr.
- 3 Ask the family members on your pedigree if they are able to roll their tongues. Write the appropriate genotypes on the chart. Note that the people on your family pedigree chart must be blood relations.

Results

Submit your family pedigree.

Analysis of results

- 1 What were the genotypes of your grandparents and parents?
- 2 How many of your family members are tongue rollers? How many are non-rollers?
- 3 What is your genotype? From which parent did you inherit this trait?
- 4 How can you justify that this trait is dominant?

Conclusion

Why is it useful to construct pedigree charts and analyse the family histories?

DIHYBRID INHERITANCE IN CORN



Background

Timing: 2 × 45 minute lessons

In the mid-1800s, Gregor Mendel experimented with how traits are passed from parents to their offspring. He discovered that **hereditary determinants (genes)** are discrete entities that do not mix in offspring. He also found that genes have different forms (e.g. **dominant** or **recessive**) and that they pass from parent to offspring according to the **laws of probability**. He chose common garden pea plants to be the focus of his research because they are easily grown in large numbers, their reproduction can be manipulated and they have obvious traits such as flower colour, seed colour, seed shape, pod colour, pod shape, flower position and plant height. Mendel selectively cross-pollinated purebred plants that had a particular trait, and observed the outcomes over many generations. A cross between plants with one pair of contrasting traits is called a **monohybrid cross**. When **pure-breeding** plants homozygous for dominant and recessive traits were crossed, the offspring of the first generation (F1) always had yellow peas. However, the following generation (F2) consistently had a 3:1 ratio of yellow peas to green peas. Mendel found similar results for other contrasting pairs of characteristics. When a hybrid plant was **test crossed**, it produced 2 phenotypes in the ratio of 1:1.

A **dihybrid cross** involves studying inheritance patterns for organisms that differ in two traits. When two hybrid plants are crossed, they produce 4 phenotypes in the ratio of 9:3:3:1. When a hybrid is test crossed with a recessive parent, it produces 4 phenotypes in the ratio of 1:1:1:1.

Aim

To examine and predict the phenotypes that will be produced in F1 and F2 generations from a dihybrid cross involving the colour and texture of corn kernels.

Hypothesis

Write a hypothesis predicting the phenotypes that will be produced in the F1 and F2 generations when a homozygous dominant parent with purple, smooth corn kernels is crossed with a homozygous recessive parent with yellow, wrinkled corn kernels.

Equipment

- corn with purple smooth kernels
- corn with yellow, wrinkled kernels
- corn of F2 generation with purple/yellow, smooth/wrinkled kernels in the ratio of 9:3:3:1
- corn produced by a test cross with purple/yellow, smooth/wrinkled kernels with a ratio of 1:1:1:1
- a pin

Method

- 1 Observe corn with homozygous dominant traits for purple colour and smooth kernels (PPSS). Handle the corn cobs carefully so that the kernels do not fall off the cob.
- 2 Observe a second corn with homozygous recessive traits for yellow colour and wrinkled kernels (ppss). Note that purple (P) is dominant to yellow (p) and smooth (S) is dominant to wrinkled (s).
- 3 Use the punnet square in Table 1 to examine the theoretical outcome of the homozygous dominant parent (PPSS) with homozygous recessive parent (ppss). Place the alleles donated by each parent in the corresponding boxes of the punnet square. Now list the possible phenotypes in the spaces below the Punnett square. Show the ratios of the phenotypes.
- 4 Use the punnet square in Table 2 to examine the theoretical outcome of the heterozygous F1 offspring (PpSs) × heterozygous F1 offspring (PpSs). Show the ratios of the phenotypes.

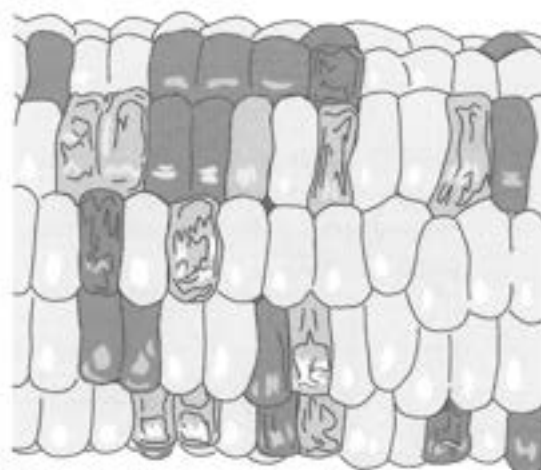


Figure 1: Test cross showing 4 phenotypes

- 5 Use the punnet square in Table 3 to examine the theoretical outcome of a test cross between heterozygous F1 offspring (PpSs) × homozygous recessive parent (ppss). Show the ratios of the phenotypes.
- 6 Obtain corn that is the result of the cross heterozygous (PpSs) × heterozygous (PpSs) for both traits. Write the four phenotypes at the tops of the columns in Table 4.
- 7 Place a pin at the end of one row on the cob. Count the colour and shape of each kernel in the row. Repeat for all the rows on the cob. (Use the pin to help you keep track of which row you started with.) One student should count the kernels and a second student should record the number of kernels for each phenotype in Table 4. Repeat the count to avoid errors.
- 8 Use the punnet square in Table 5 to calculate the ratios for the cross.
- 9 Compare your results with the theoretical results you obtained for that cross.
- 10 Collate the class results of all the phenotypes and calculate the ratios. Compare your results with the class results and note any differences in phenotype ratios.
- 11 Use the punnet square in Table 6 to work out the phenotypes that will be produced when a heterozygous corn from F1 (PpSs) is crossed with a homozygous recessive parent (ppss). Note that if an individual's genotype is heterozygous, the dominant trait will be expressed in the phenotype. There are four possible phenotypes for the offspring of this cross, and both traits are in each phenotype.

Theoretical results

Table 1 Homozygous purple smooth (PPSS) × homozygous yellow wrinkled (ppss)

	PPSS	
ppss	PS	PS
	ps	ps

Phenotypes: _____

Ratio: _____:_____:_____:_____

Table 2 Heterozygous purple smooth (PpSs) × heterozygous purple smooth (PpSs)

		PpSs			
		PS	Ps	pS	ps
PpSs	PS				
	Ps				
	pS				
	ps				

Phenotypes: _____

Ratio: _____:_____:_____:_____

Table 3 Test cross: heterozygous F1 purple smooth offspring (PpSs) × homozygous yellow wrinkled recessive parent (ppss)

	PpSs			
	PS	Ps	pS	ps
ps				

Phenotypes: _____

Ratio: _____:_____:_____:_____

Observed results

Table 4

Phenotype	Purple, smooth	Purple, wrinkled	Yellow, smooth	Yellow, wrinkled
Number				
Ratio				

Table 5 Heterozygous purple smooth (PpSs) × heterozygous purple smooth (PpSs)

	PpSs				
	PS	Ps	pS	ps	
PpSs					
	PS				
	Ps				
	pS				
	ps				

Phenotypes: _____

Ratio: _____:_____:_____:_____

Table 6 Test cross: heterozygous F1 purple smooth offspring (PpSs) × homozygous recessive yellow wrinkled parent (ppss)

	(PpSs)				
	PS	Ps	pS	ps	
ppss					
	ps				
	ps				

Phenotypes: _____

Ratio: _____:_____:_____:_____

Analysis of results

- 1 What is meant by 'dihybrid ratio'?
- 2 Which alleles are dominant and which alleles are recessive for the colour and the shape of corn kernels?
- 3 Which colour/shape combination appeared most often in the F2 generation? Which appeared least often?
- 4 What phenotypes were produced in the F1 offspring when pure-breeding purple smooth corn was crossed with pure-breeding yellow, wrinkled corn?
- 5 What phenotypes were produced in the F2 offspring when two F1 heterozygotes were crossed? What were the phenotype ratios?
- 6 What phenotypes were produced when a F1 heterozygote was crossed with a homozygous recessive parent during test crossing? What were the phenotype ratios?
- 7 Did your results support your hypothesis?

Conclusion

- 1 How is a monohybrid cross different from a dihybrid cross?
- 2 What is the advantage of using punnet squares?

Background

Timing: 2 × 45 minute lessons

Evolution can be defined as the change in the frequency of one or more alleles in a **population's gene pool** from one generation to the next, due to natural selection. There are three main types of evolution: **divergent**, **convergent**, and **parallel**.

In convergent evolution, unrelated species become increasingly similar in appearance because of a shared environment. Convergent evolution results in **analogous** structures that carry out a similar function but are not similar in basic structure. This is because they are not inherited from a common ancestor. For example, sharks, porpoises, and penguins are not related through a common ancestor, but they all have peripheral fins and bodies that are similar in shape. These traits are a direct result of adapting to aquatic life. Similarly, wings developed independently in insects, reptiles, birds and bats.

In divergent evolution, two or more species develop from a common ancestor. This happens through adaptation to different environments. Divergent evolution results in **homologous** structures. These are shared by related species because they have been inherited from a common ancestor. For example, the forearms of various vertebrates are homologous because they have similar bone structures but different functions.

Parallel evolution is the development of the same characteristic in unrelated organisms through adaptation to similar environmental conditions. For example, the marsupial mammals of Australia are similar to the placental mammals from other parts of the world. However, marsupials in Australia evolved in isolation after Australia separated from the other continents.

Aim To analyse how different types of evolution give rise to homologous and analogous structures.

Equipment • diagrams and specimens of homologous structures and analogous structures • paper and coloured pencils

Part A – Homologous structures due to divergent evolution

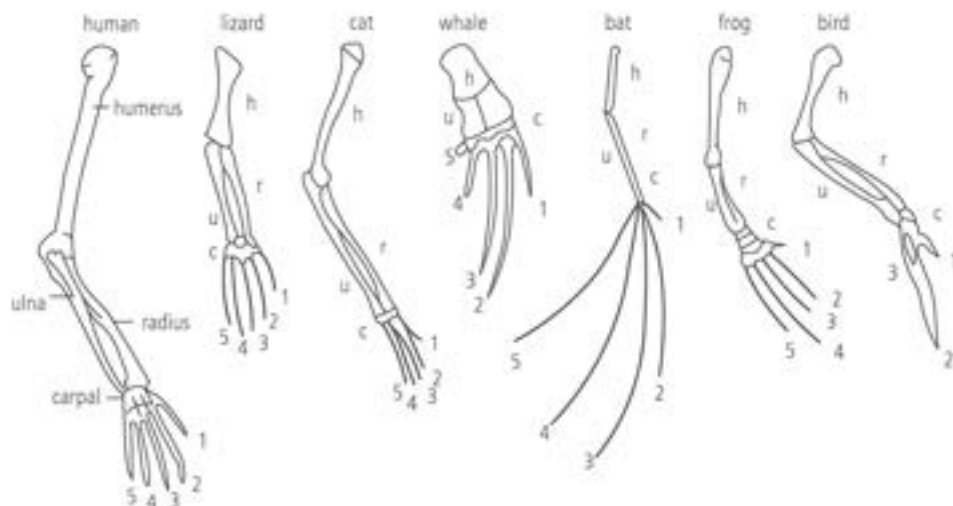


Figure 1: Homologous structures

Method

- 1 Colour the homologous bones in Figure 2. Use green for the humerus, red for the radius, blue for the ulna, yellow for the carpals and metacarpals, purple for the phalanges.
- 2 Compare the bones in the human forearm with the bones in the forearms of the other vertebrates in Figure 2. Note the similarities and differences in bone structure, including the relative size and thickness of the bones. Record your results in Table 1.

Results

Record the size and thickness of each bone in Table 1.

Table 1 Homologous bone structures

Bone structure and function	Human (mammal)	Lizard (reptile)	Cat (mammal)	Whale (mammal)	Bat (mammal)	Frog (amphibian)	Bird (aves)
Humerus							
Radius							
Ulna							
Metacarpals							
Habitat							
Function							

Analysis of results

- 1 What are homologous structures?
- 2 Do homologous structures have the same functions in different organisms?
- 3 What do homologous structures have in common?
- 4 Do homologous structures evolve due to convergent or divergent evolution? Explain.

Part B – Analogous structures due to convergent evolution

The spiny anteater of Australia, the giant pangolin of Africa, the giant anteater of Latin America and the giant armadillo of North America all have long, sticky, worm-like tongues; long, hairless snouts; no teeth; and sharp claws. Despite these similarities, these creatures are unrelated. (In fact, the spiny anteater lays eggs and is a close relative of the duck-billed platypus.) These four types of anteaters have evolved similar adaptations because they occupy similar niches – dining on ants and termites, hunting in high grass and swimming in the dark.

The last common ancestor of flying reptiles, birds and mammals did not have wings, although it certainly had a forelimb. The wing structures in these creatures formed independently. They are analogous structures because they have similar functions.



Figure 2: Anteaters from different continents

The pterodactyl wing was supported by an elongated fifth digit. It had modified epidermal structures that were wing-supporting fibres, and others that possibly formed hair-like structures to provide insulation. The bird wing is primarily supported by an elongated radius, ulna and modified wrist bones. It also has feathers. The insect wing is composed of transparent non-living chitin. This spans the spaces between a network of wing veins that fan out into the wing shape. The bat wing is a membrane. It is supported by the arm and the greatly elongated fingers of the hand, which support the distal part of the wing.

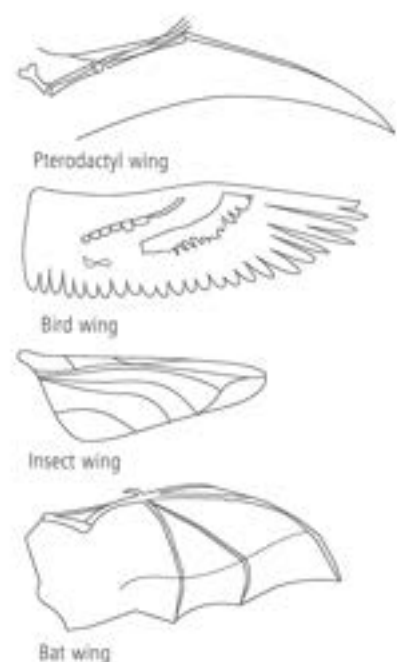


Figure 3: Wings of a pterodactyl, bird, insect and bat

The dolphin and the ichthyosaur look alike but they are very different species. Because they both live in an aquatic habitat, both have evolved a streamlined shape that moves through water with minimal resistance.

Method

- Examine Figures 4 and 5 and the information above. In Table 2, record the similarities and differences among the wings of the pterodactyl, bird, insect and bat, and between the dolphin and the ichthyosaur.

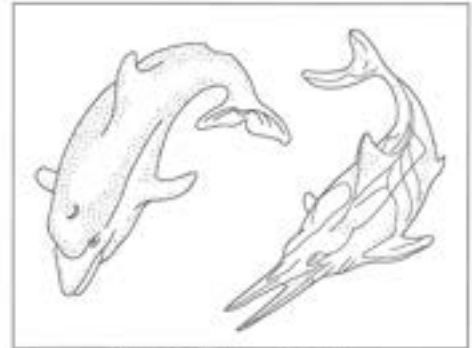


Figure 4: Dolphin and ichthyosaur

Results

Table 2

Features and function	Pterodactyl (reptile)	Bird (aves)	Bat (mammal)	Insect (arthropod)	Dolphin (mammal)	Ichthyosaur (reptile)
Body shape, structures present						
Habitat						
Function						

Analysis of results

- Why do all anteaters look similar even though they are unrelated?
- Does the dolphin and the ichthyosaur have a common ancestor?
- Why do the dolphin and the ichthyosaur resemble each other so closely even though the dolphin is a mammal and ichthyosaur is a reptile?
- What type of evolution results in analogous structures?
- Why are the wings of insects and birds analogous and not homologous?
- What data do you need to determine if two structures are homologous or analogous?

Part C – Parallel evolution in Australian marsupials and placentals from other continents

In parallel evolution, two organisms will often possess similar adaptations, but the embryology of the structures clearly shows that there is little or no underlying structural similarity. (That is, there is similarity without a common evolutionary origin.) These adaptations are said to occur in parallel.

Marsupials in Australia and placental mammals in North America have adapted in similar ways to a particular food supply and climate. They separated from a common ancestor more than 100 million years ago, and each species continued to evolve independently. They are similar in overall shape, mode of movement, feeding and foraging. The marsupials have undergone an adaptive radiation to occupy the diversity of habitats in Australia, just as the placentals have radiated across North America. The lack of carnivorous placental mammals in Australia allowed for reproductive isolation and random mutation to take over. This created a species that is unique to the Australian continent.

Marsupial young are born relatively underdeveloped. In some but not all marsupial species, the mother develops a special pouch on her body in which to nurse her young. In placental mammals, the young develop in the uterus, nourished by a specialised organ attached to the uterus wall – the placenta.

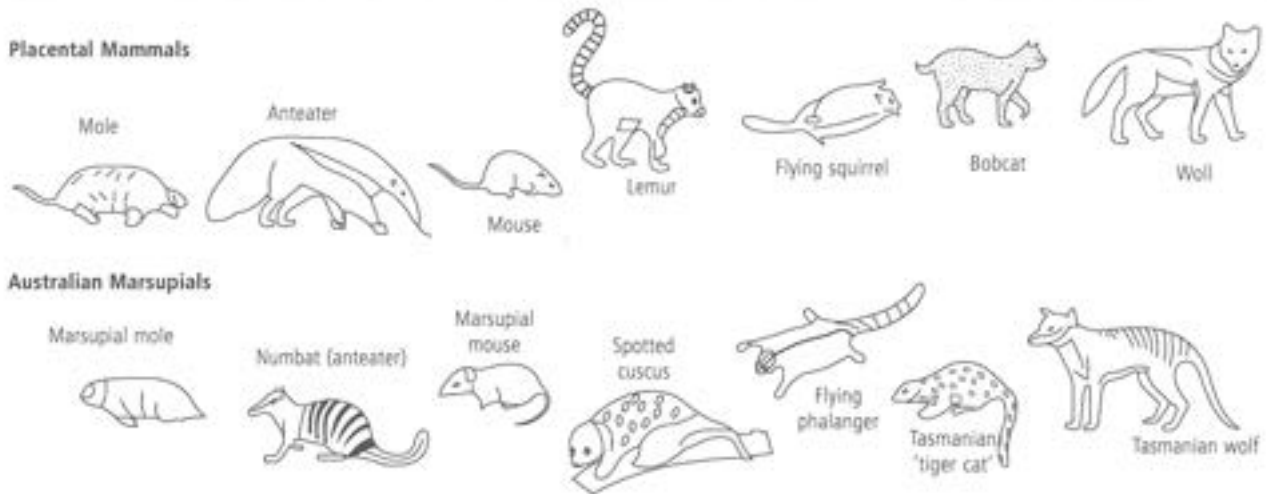


Figure 5: Australian marsupials and placentals from other continents

Method

- 1 Examine Figure 5 and the information on the previous page. In Table 3, record the similarities and differences between placental mammals and Australian marsupials.

Results

Table 3

Similarities and differences	Placentals	Marsupials
Similarities		
Differences		
Functions		

Analysis of results

- 1 In what ways do Australian marsupials and placental mammals look similar? Explain why this occurred.
- 2 Name the type of evolution that leads to two unrelated species having similar appearances and adaptations.
- 3 Why did adaptive radiation occur in marsupials in Australia?
- 4 How do marsupials differ from placental mammals?
- 5 Identify the types of evolution shown in Figure 6.

Conclusion

- 1 What are the differences between divergent, convergent and parallel evolution?
- 2 Is it homologous structures or analogous structures that show that organisms are more closely related? Explain.

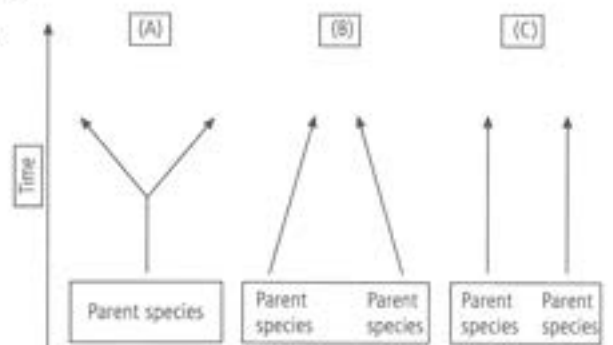


Figure 6: Types of evolution

NATURAL SELECTION IN PEPPERED MOTHS



Background

Timing: 2 × 45 minute lessons

The peppered moth, *Biston betularia*, occurs in two forms: f. typica (light form) and f. carbonaria (dark form). 150 years ago, the species consisted almost entirely of 'typical' forms. They had predominantly light grey scales interspersed with black scales. This gave the moth the name 'peppered'. The **Industrial Revolution** in Britain began in the 1840s. During this time, a large amount of airborne **pollution** was produced, in the forms of soot and toxic gases. This pollution killed the lichen on the bark of trees growing in industrialised areas.

By the 1950s, at least 90% of peppered moths living in polluted areas were dark in colour. The melanism shown by peppered moths is known as **industrial melanism**. This is because it is caused by the soot from industry. Researchers noted that the increase of the proportion of dark moths paralleled the increase in industrial pollution. They hypothesised that the dark moth was better camouflaged than the light moth when they settled on soot-darkened tree trunks. These observations suggested that the light moths were removed from the population by birds because they were more conspicuous than the dark moths. Additional support for this hypothesis came from non-industrial regions where the light moths greatly outnumbered the dark moths. In these areas, the selection was reversed and the light moths had the advantage in hiding from birds on the lichen-covered tree trunks.

The response of the peppered moth to industrial pollution in England is a well-known example of **natural selection**. The **Hardy-Weinberg Law** provides a mathematical formula for calculating the frequencies of alleles and genotypes in populations. According to this law, both gene and genotype frequencies will remain unchanged in equilibrium unless outside forces change those frequencies. The Hardy-Weinberg Law depends on several basic assumptions: an infinite, randomly-mating population, with no mutation, migration or selection within the population.

Aim

To analyse experiments conducted by biologists on peppered moth populations in England during the Industrial Revolution.

Equipment • paper and pen

Hasebroek (1926) proposed that salts of lead and manganese present in the airborne pollutant particles caused the mutation of the genes for melanin production. He did an experiment to test his theory. Hughes McKenney (1932) and Thomasen and Lemche (1933) conducted similar experiments but obtained different results.

Bernard Kettlewell believed that industrial melanism was due to selective predation by birds. In 1955, he used three experimental approaches to test his theory. The following table shows how the experiments were conducted and the results obtained.



Biston betularia betularia f. typica
– light moths (aa)



Biston betularia betularia f. carbonaria
– dark moths (AA or Aa)

Figure 1: Dark and light moths

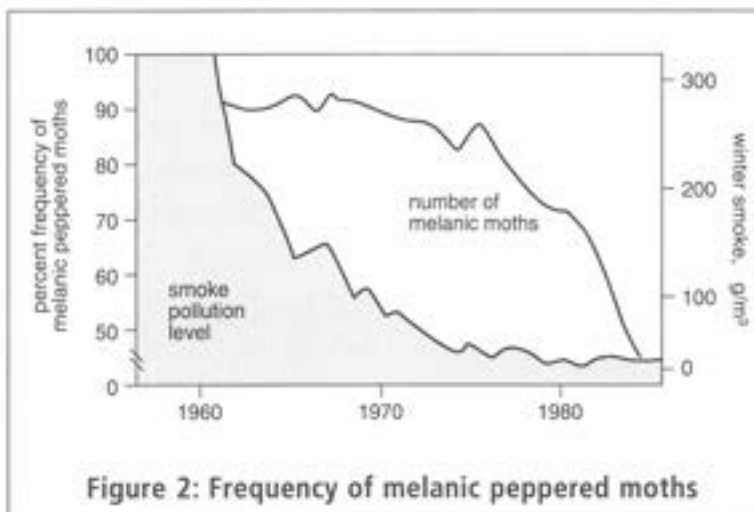


Figure 2: Frequency of melanic peppered moths

Experiments	Results	Percentage of moths captured/inference
Hasebroek fed larvae of white moths with polluted leaves.	8% were dark moths and 92% were light moths.	Dark moths: _____% Light moths: _____% Inference: _____
McKenney, Thomasen and Lemche fed larvae of white moths with polluted and unpolluted leaves.	8% were dark moths and 92% were light moths.	Dark moths: _____% Light moths: _____% Inference: _____

Kettlewell's experiments

Aviary: He released moths into a large aviary containing a pair of great tits (<i>Parus major</i>).	All the moths were eaten by the birds.	Moths captured.
Unpolluted forest: 473 dark moths and 496 light moths were marked and released in an unpolluted forest. They were later recaptured and the marked ones were counted.	Out of 473 dark moths, 30 were recaptured. Out of 496 light moths, 62 were recaptured.	Dark moths: _____% Light moths: _____% Ratio: _____ Inference: _____
Polluted forest: 447 dark moths and 137 light moths were marked and released in an unpolluted forest. They were later recaptured and the marked ones were counted. Note: A spot of white paint was placed on the underside of the moths.	Out of 447 dark moths, 123 were recaptured. Out of 137 light moths, 18 were recaptured.	Dark moths: _____% Light moths: _____% Ratio: _____ Inference: _____

Analysis of results

- How did the experimental results of McKenney, Thomasen and Lemche differ from the results of Hasebroek?
- Why did Kettlewell paint the white spot on the underside of the moths?
- Why could Kettlewell recapture only a small percentage of the marked moths?
- Suggest a likely explanation for the difference in Kettlewell's results for the polluted forest and the unpolluted forest.
- Following the passage of anti-pollution legislation in the 1950s, industrial melanism began to decline. The percentage of dark moths dropped from 93% in 1959 to 90% in 1962. By 1984, the percentage of melanic decreased to 61% and in 1996 the percentage was only 8.5%. Suggest a likely explanation for this.
- What did Kettlewell conclude from his experiments?

Part B – Applying Hardy-Weinberg Law to changes in allele frequency in moth population

Aim To use the Hardy-Weinberg Law of Equilibrium to investigate changes in allele frequencies in moth populations due to natural selection.

Hardy-Weinberg formula

- Let p represent the frequency of one gene in the pool and q the frequency of its single allele.
- So, $p + q = 1$
- p^2 = the fraction of the population homozygous for p
- q^2 = the fraction of the population homozygous for q

- $2pq$ = the fraction of heterozygotes
- $(p + q)^2 = p^2 + 2pq + q^2$

Method

- 1 Let AA be the genotype of the homozygous dark moths. Let aa be the genotype of the homozygous light moths. Let Aa be the genotype of the heterozygous dark moths.
- 2 The frequencies of alleles A and a during and after pollution are given below.

Alleles	During pollution	After pollution
p = A	0.6	0.5
q = a	0.4	0.5

- 3 Use the Hardy-Weinberg formula to calculate the frequencies of alleles before and after pollution by applying the following equation.
 $(p + q)^2 = p^2 + 2pq + q^2 = (A + a)^2 = a^2 + 2Aa + A^2$
- 4 Use the punnet squares under Results to show the changes in frequencies.
- 5 Given that the total number of moths was 100, calculate the number of dark and light moths.

Results

Analysis of results

During pollution		
If q = 0.4 p = 1 - 0.4 = 0.6		
	p = 0.6	q = 0.4
p = 0.6		
q = 0.4		

$(p + q)^2 = (0.6 + 0.4)^2 =$
 Homozygous dark moths = _____
 Heterozygous dark moths = _____
 Total dark moths = _____
 Total light moths = _____

After pollution		
If q = 0.5 p = 1 - 0.5 = 0.5		
	p = 0.5	q = 0.5
p = 0.5		
q = 0.5		

$(p + q)^2 = (0.5 + 0.5)^2 =$
 Homozygous dark moths = _____
 Heterozygous dark moths = _____
 Total dark moths = _____
 Total light moths = _____

- 1 What is meant by an allele?
- 2 Which is the dominant allele and which is the recessive allele?
- 3 From your results, did you find any changes in the frequencies of dark moths and light moths during and after pollution?
- 4 Name the process that could explain how the change in the Biston betularia populations in England came about.
- 5 Is it likely that any dark-coloured moths existed in the natural population before the Industrial Revolution? Explain.

Conclusion

- 1 What conclusion can you draw from the experiments described above?
- 2 What can you conclude about the changes in allele frequencies in moth population during and after pollution in Britain?

ADAPTIVE RADIATION OF FINCHES AND TORTOISES IN GALAPAGOS ISLANDS



Background

Timing: 2 × 45 minute lessons

The Galapagos Islands consist of 13 major islands. They are located in the Pacific Ocean, 1000 km from the coast of Ecuador which is in South America. **Charles Darwin's** visit to the Galapagos Islands in 1835 was the starting point for his theory of **natural selection**. This theory describes a process by which organisms that have favourable variations survive and reproduce at a higher rate than other organisms.

An inherited variation that increases an organism's chance of survival in a particular environment is called an **adaptation**. The finches and the giant tortoises living on the Galapagos Islands are two examples of **adaptive radiation**. This means that a single lineage has diversified into a variety of species that have different adaptive features. Adaptive radiation is an example of **divergent evolution**.

Approximately 10 000 years ago, small **finches** (grass quits) travelled 1000 km across the Pacific Ocean from South America to the Galapagos Islands. These islands were relatively free of competitors. The grass quits underwent radiation, producing 13 species that occupy a variety of niches.

10 million years ago, the ancestors of the giant **tortoises** arrived in the Galapagos Islands by clinging to a piece of wood that drifted from a river mouth on the Pacific coast of the South American mainland all the way to San Cristobal Island. The tortoises then spread throughout the islands. When Darwin arrived, there were approximately 300 000 giant tortoises on 11 of the islands. The tortoises on individual islands developed into their own sub-species. This was due to the effect of the **founder principle, genetic drift and natural selection**. These islands were isolated and the animals were subjected to different environmental conditions, which resulted in their gradual evolution into different species.

Aim

To analyse the information provided to understand how adaptive radiation occurred in giant tortoises and finches on the Galapagos Islands.

Equipment

- photographs and information about tortoises, finches and cacti from the Galapagos Islands
- map of Galapagos Islands

Part A – Adaptive radiation in Galapagos tortoises



Figure 1: Dome-shaped tortoise



Figure 2: Saddle-backed tortoise

The Galapagos Islands once had 15 sub-species of giant tortoises. Today, 11 sub-species exist. Only 15 000 tortoises are left in these islands because of excessive hunting by humans and introduced predators. The original ancestors of the tortoises were probably much smaller in size and evolved into the present-day giants after their arrival in the Galapagos Islands. This is due to a phenomenon seen in many island ecosystems, where gigantism evolves because there is no longer any need to hide from predators or to compete with similar animals for food. There are three types of Galapagos tortoises, distinguished by the shape of the shell, size, colour and behaviour. These animals have developed different characteristics to suit the environments they live in and the types of food available.

Saddle-backed tortoises (*Geochelone elephantopus*) are mostly found on the lower, drier islands where food is scarce and a smaller body is an advantage. The high opening of the shell allows them to raise their necks higher to reach the sparse vegetation. This is also helped by their longer necks and legs. They feed on plants that are mostly above their heads, such as the tree-like *Opuntia* (prickly pear) cacti.

Dome-shaped tortoises (*Geochelone gigantea*) are found on the upper parts of the islands, where plant growth is dense and thick. They have round-shaped shells and very short necks and limbs. They feed on grasses and low-lying shrubs.

Intermediate tortoises are a mix between the saddle-backed tortoise and the dome-shaped tortoise.

Galapagos cacti provide the habitat and food source for tortoises. Giant tortoises feed on the prickly pear pads of *Opuntia*. By doing this, they exert a selection pressure for cacti with defence mechanisms that prevent them from being eaten. This pressure has favoured the tall, tree-like *Opuntia echios* that has hard bark and spines. This cacti is more common on islands that have tortoises, whereas the shrub form *Opuntia helleri* is more common on islands that have never had tortoises.

Coevolution is the mutual evolutionary influence between two species that become dependent on each other. Each party in a co-evolutionary relationship exerts selective pressures on the other, thereby affecting each others' evolution. Coevolution is an extreme example of mutualism. Giant tortoises feed on the prickly pear pads of *Opuntia*. In doing so, they exert a selection pressure for cacti with defence mechanisms to prevent being eaten. This pressure has favoured the tall, tree-form *Opuntia* with hard bark and spines on islands with tortoises, whereas the shrub form is more common on islands that have never had tortoises.

Method

- 1 Examine Figures 1 and 2 and the information above. Then fill in Table 1.

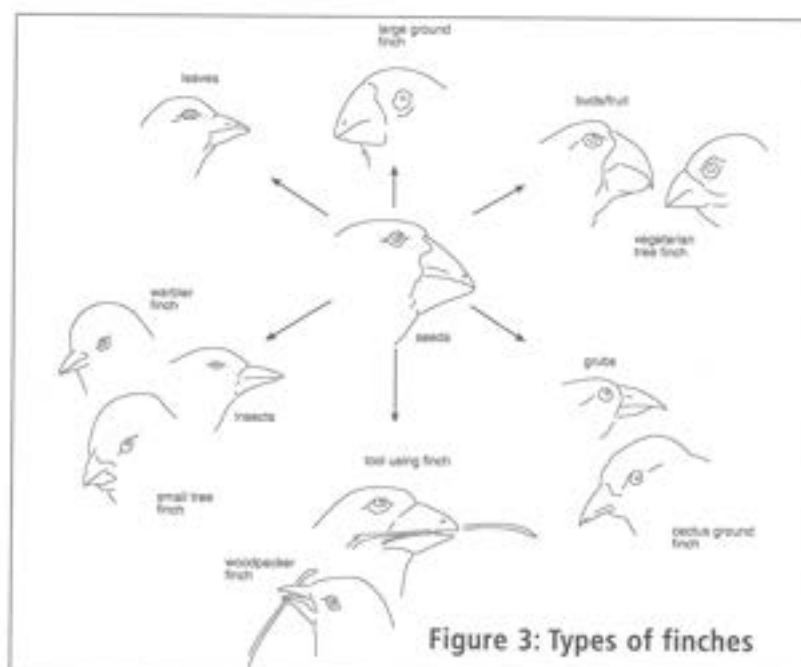
Results

Table 1

Characteristics	Saddle-backed tortoise	Dome-shaped tortoise
Type of shell		
Type of neck and legs		
Type of environment		
Type of food eaten		

Analysis of results

- 1 How did the tortoises reach the Galapagos Islands from the South American mainland? How long ago did they arrive?
- 2 How many sub-species exist today? Explain why this number has decreased.
- 3 Why did the size of the tortoises increase in these islands?
- 4 What are the differences between saddle-backed and dome-shaped tortoises?
- 5 Which selective pressures led to the tortoises evolving into these two types?
- 6 What is coevolution? Explain how the cacti and the tortoises coevolved.

Part B – Adaptive radiation in Darwin's finches**Figure 3: Types of finches**

The main differences between finches in the Galapagos Islands are in body size and the size and shape of their beaks. On different islands, finch species have adapted to different diets: seeds, insects, flowers, the blood of seabirds and leaves. There are 13 species of Darwin's finches: small ground finch, medium ground finch, large ground finch, vegetarian finch, woodpecker finch, mangrove finch, small tree finch, medium tree finch, large tree finch, cactus finch, large cactus finch, sharp-billed ground finch and the warbler finch. Each species has a different shaped beak for feeding on foods such as insects, flowers, leaves and seeds. One species, the sharp-billed ground finch (also known as the vampire finch), feeds on blood from other birds. Adaptive radiation has occurred in these birds in order to adapt to different environments and the availability of food.

Method

Examine Figure 3 and the information above. Then fill in Table 2.

Results**Table 2**

Types of finches	Food eaten	Types of bills
Tree finches		
Warbler finches		
Ground finches		

Analysis of results

- 1 From where did the ancestral finches come to Galapagos Islands?
- 2 What is the difference between a species and a sub-species?
- 3 How many species of finches exist in Galapagos Islands today? How do they differ from one another?
- 4 Why do finches have beaks of different shapes?
- 5 How could two or more species of finches co-exist in the same habitat without competing?
- 6 What selective pressures acted on finches to lead to adaptive radiation?

Conclusion

- 1 Explain what occurred to the finches and tortoises living on Galapagos Islands.
- 2 Name the process that led to divergent evolution in these two species.

COMPARISON OF PRIMATE FEATURES



Background

Timing: 2 × 45 minute lessons

Primates evolved about 30 million years ago in Africa. One branch of primates evolved into the old world monkeys and the new world monkeys. The other branch evolved into **hominids**. The hominids includes humans, extinct species of human-like creatures and the great apes: chimpanzees (genus *Pan*), gorillas and orangutans (genus *Pongo*). It has been estimated that humans diverged from gorillas about 8 million years ago, and from chimpanzees about 5 million years ago.

Biologists believe that chimpanzees are our closest evolutionary relatives because we have 99.4% of DNA in common with them. The earliest genus of hominids is the **Australopithecus**, meaning 'southern ape'. About 2 million years ago, as many as 6 hominid species evolved in response to climate changes associated with the beginning of the Ice Age. Two major groups developed: the Australopithecines and the line that led to the **genus Homo**. The Australopithecines were generally smaller brained and non-users of tools. They died out 1 million years to 2–2.5 million years ago.

The line that led to the genus *Homo* was larger brained and were makers and users of tools. **Homo habilis** evolved with a larger brain and different shaped skull and teeth. About 1.8 million years ago, **Homo erectus** evolved and appears to have migrated from Africa into Europe and Asia. *H. erectus* had a larger brain, flatter face and prominent brow ridges. *H. erectus* is similar to modern humans in size, but there are differences in the shape of the skull, a receding chin, brow ridges and differences in teeth. Between 100 000 and 500 000 years ago, *H. erectus* disappeared and was replaced by a new species, **Homo sapiens**.

According to Donald Johanson (who discovered the skeleton of Lucy in Africa) *Australopithecus afarensis* is the ancestor of both *A. africanus* and modern humans because they had a combination of human bipedalism and ape-like features. Louis Leakey did not agree with Johanson. He believed that there must be another species which gave rise to both the *Homo* species and the *Australopithecine* species.

Aim To compare the features of different primate species.

Equipment • Diagrams and models of skeletons, skulls, pelvic girdles, feet and hands of a chimpanzee, gorilla, *Australopithecus afarensis*, *Australopithecus africanus* and *Homo sapiens sapiens*

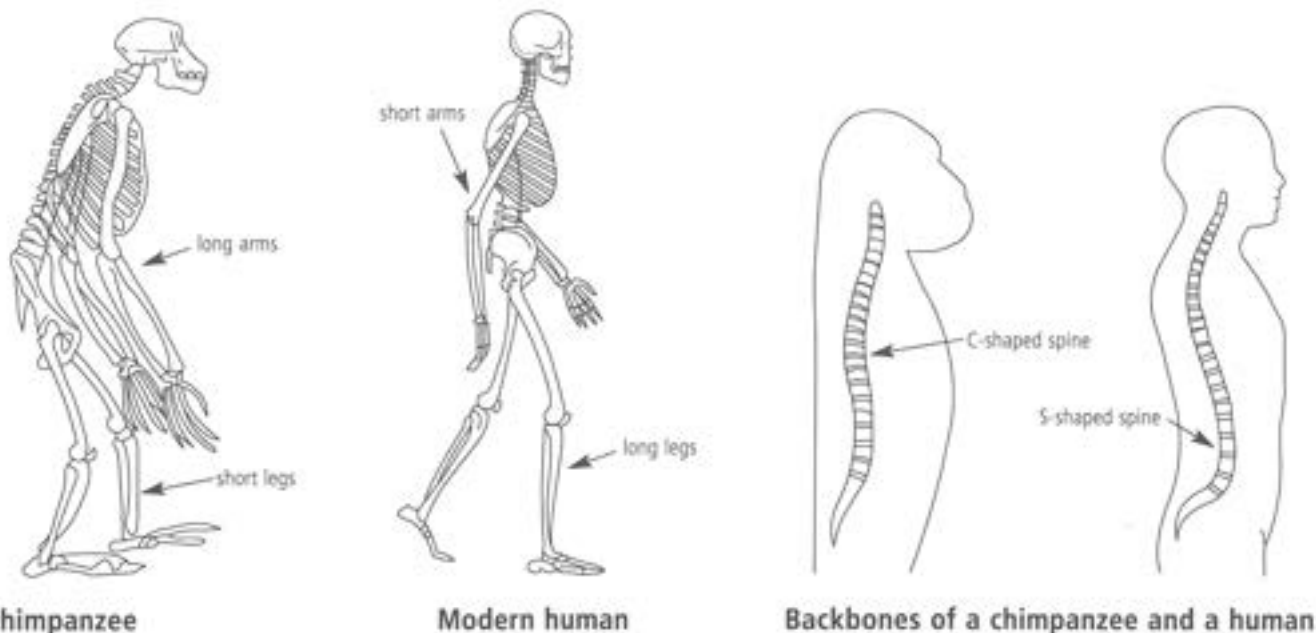


Figure 1: Skeletons and backbones of a chimpanzee and *H. sapien*

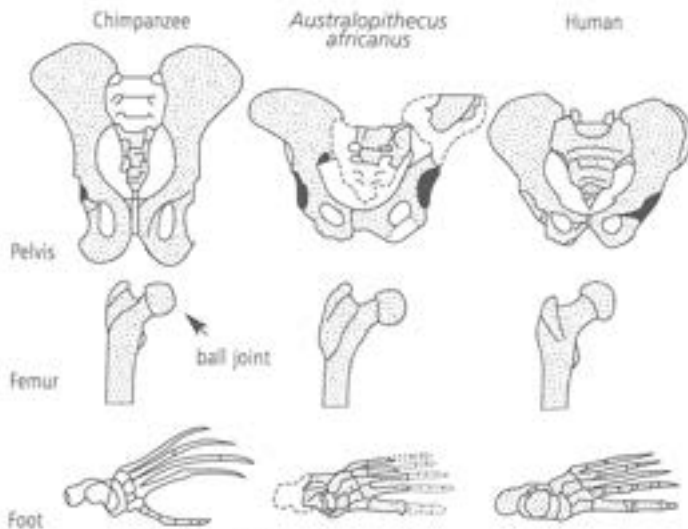


Figure 2: Pelvic girdles, femur bones and feet of a chimpanzee, *A. africanus* and human

Chimpanzee: The foot appears much more hand-like and has an opposable toe which helps in grasping branches and knuckle walking.

***A. afarensis*:** The foot has no opposable toe and the phalanges are curved and proportionally longer than in humans. This enables it to walk on two feet (bipedal).

***H.sapiens*:** The foot has no opposable toe. The phalanges are long and straight, which enables it to walk on two feet.



Hands of a *Australopithecus*: opposable thumb, phalanges are slightly curved



Hands of a gorilla: opposable thumb, the phalanges are adapted for knuckle walking



Hand of *H. sapiens*: opposable thumb, the phalanges are curved for manipulation

Figure 3: Hands of a gorilla, *Australopithecus* and *H. sapiens*

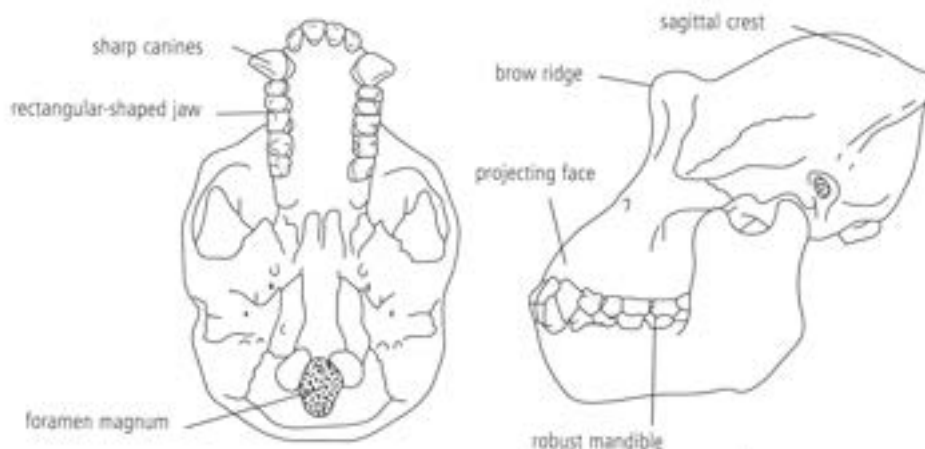
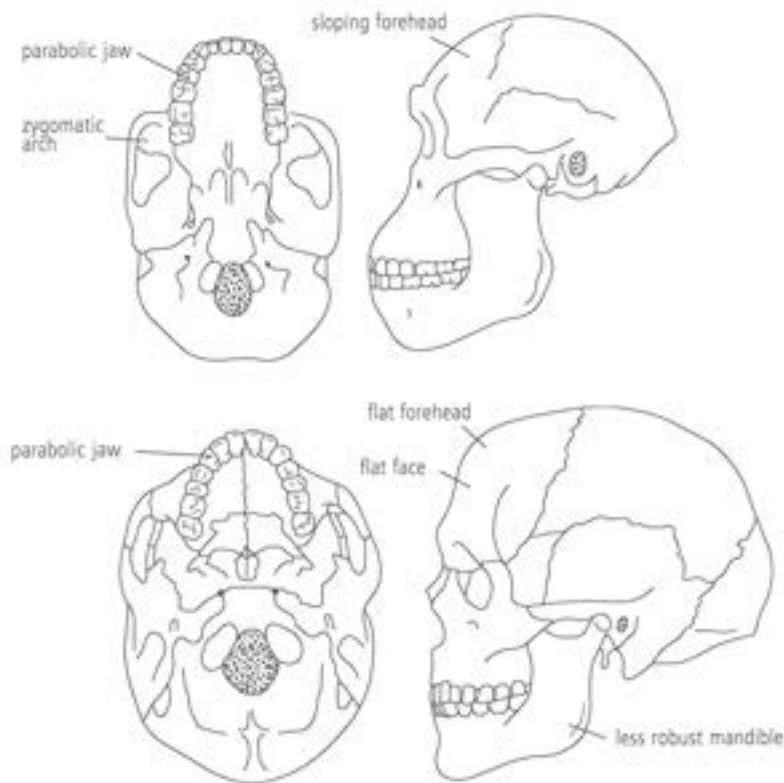


Figure 4: Skull and jaw of a female gorilla

The dark area at the bottom of the skull is the foramen magnum. The area where the spine enters the cranium is at the back of the skull. The molar teeth are very large, their canines are huge and tusk-like, but they have small incisors. Their snouts are longer and prognathic, and very large sagittal crest is present. They have a small cranium and a strong brow ridge (brow bone). They have a concave face, sloping forehead, heavy jaw and no chin.



Australopithecines: The foramen magnum has moved markedly forward towards the centre. This indicates a more upright posture and bipedal locomotion. The jaw is less heavy and parabolic in shape like that of *H. Sapiens*. The teeth are similar to those of humans. The molars are similar, although larger. They do not have the large canine teeth of apes. They have sloping forehead, a bony ridge over the eyes, a flat nose, and no chin.

The skull of ***Homo sapiens sapiens*** has a well-developed chin, large cranium, small parabolic shaped jaw, small canines, very small brow ridge, strong cheekbones and no sloping of the forehead.

Figure 5: Skulls and jaws of, *A. africanus* and *H. sapiens*

Method

- 1 Examine Figure 1. In Table 1, record the differences between the backbone and skeleton of a chimpanzee and a modern human.
- 2 Examine Figure 2. In Table 2, record the differences between the pelvic girdles, femur and feet of a chimpanzee, *A. africanus* and a modern human.
- 3 Examine Figure 3. In Table 2, record the differences in the shape and size of the fingers and toes of a gorilla (a knuckle walker), *A. afarensis* (bipedal) and *H. sapien* (bipedal).
- 4 Examine Figure 4. In Table 3, record the differences in the skulls, teeth and jaws of a gorilla, *A. africanus* and *H. sapiens*.

Results

Table 1 Differences between the skeletons of a chimpanzee and an *H. sapien*

Features	Chimpanzee	<i>H. sapiens</i>
Backbone: shape		
Rib cage : size and shape		
Skull: size and attachment		
Arms and legs: length and use		

Table 2 Differences between the pelvis, femur, hands and feet of a gorilla, Australopithecus and H. sapien

Features	Chipanzee	Australopithecus	H. sapien
Pelvis: size and shape			
Femur: shape			
Fingers in the hand			
Opposable thumb			
Type of locomotion			
Toes in the feet			

Table 3 Differences in skulls, jaws and teeth of gorilla, Australopithecus and H. sapien

Features	Gorilla	Australopithecus	H. sapiens
Skull: shape and size			
Brow ridge: size			
Sagittal crest			
Foramen magnum (the hole below the skull): position			
Jaw: shape and size			
Forehead and chin: shape and size			
Teeth: type and size			

Analysis of results

- 1 Which of the human skeletal traits you examined are related to walking upright?
- 2 During human evolution, which trait is more likely to have evolved first: bipedalism or expansion of brain size?
- 3 Refer to Table 4 to answer the following questions.
 - a Which species has the smallest brain? Largest brain?
 - b What is the difference between Homo brains and Australopithecine brains?
 - c Draw a line graph of the volume of the brain against the primate.

Table 4 Size of the brain

	Latest date (millions of years)	Brain volume (cm ³)
Gorilla	0	505
A. afarensis	2.8	384
A. africanus	2.3	420
P. boisei	1.3	488
P. robustus	1.0	502
H. habilis	1.6	597
H. ergaster	1.5	804
H. erectus	0.3	950
H. sapiens	0	1350

Conclusion

- 1 Summarise the changes that have occurred during human evolution.
- 2 Why did bipedalism develop before the increase in brain size? Explain.

SUGGESTED ANSWERS



Extraction of DNA from onions

Analysis of results

1 The onion and detergent/salt solution were blended to break up the cell membrane and the nuclear membrane. 2 Meat tenderiser was added to eliminate any protein contamination. 3 When ethanol was poured in, DNA separated and floated in the alcohol. 4 The DNA obtained from the onion cells looks like white mucus. 5 The phenol red indicator turned the DNA pink because it contains nucleic acids.

Conclusion

1 The cells of all living organisms contain DNA. 2 DNA is mainly located in the nucleus of the cell. Mitochondria and chloroplasts also contain DNA. 3 DNA controls the hereditary characteristics of organisms. It also controls protein synthesis.



Modelling a DNA molecule and DNA replication

Part A – Modelling a DNA molecule

Analysis of results

1 Chromosomes are thread-like structures found in the nucleus of a cell. They are made up of deoxyribonucleic acid (DNA) and proteins. 2 Genes are sections of chromosomes which code for proteins and inherited instructions that control structure and function. 3 The sequence of bases on one side of the DNA determines the sequence of bases on the opposite strand. 4 If one strand has GGACTGTTA, the opposite strand will have CCTGACAAT.

Part B – Modelling DNA replication

Analysis of results

1 After replication, the two newly formed strands are identical to each other; but one strand is the original strand and the second strand is the new strand. This is known as the semi-conservative model. 2 A three-dimensional model of DNA could be made with Lego blocks or beads of different colours and pipe cleaners.

Conclusion

1 Deoxyribonucleic acid or DNA is made up of sub-units called nucleotides. A nucleotide consists of one nitrogen base, one phosphate group and one deoxyribose molecule. There are four different bases: adenine, cytosine, guanine and thymine. Adenine always pairs with thymine and cytosine always pairs with guanine. The nucleotides make up two chains that are linked and twisted around one another in the form of a double helix. 2 During DNA replication, DNA helicase enzyme moves up along the DNA ladder, unzipping the molecule as it moves along. New nucleotides move in to each side of the unzipped ladder. The bases on these nucleotides are very particular about what they connect to. Cytosine (C) will "pair" to guanine (G), and adenine (A) will "pair" to thymine (T). Two identical molecules of DNA are left behind. Each contains one side of the original DNA and one side made up of "new" nucleotides. The two new double-stranded chains then twist into two separate double helices. This forms two identical DNA molecules.



Modelling protein synthesis

Results

DNA template strand	mRNA sequence	tRNA anticodons	Amino acids
A C G	U G C	A C G	Cys
T T C	A A G	U U C	Lys
C C G	G G C	C C G	Gly
G C G	C G C	G C G	Arg
T G C	A C G	U G C	Thr

Analysis of results

1 Transcription takes place in the nucleus. 2 A DNA molecule consists of two strands of nucleotides twisted together to form a double helix. RNA consists of one strand of nucleic acids. The sugar is deoxyribose in DNA and ribose in RNA. Adenine and guanine are both found in DNA and RNA.

3 Modelling protein synthesis (cont'd)

Cytosine is found in both DNA and RNA, but thymine is only found in DNA and uracil is only found in RNA. **3** The sense strand acts as the template strand for mRNA transcription. **4** rRNA make up ribosomes, tRNA transport amino acids to ribosomes and mRNA, copied from DNA, conveys information from chromosomes to ribosomes. **5** Anticodon is a specific sequence of three nucleotides in tRNA. They are complementary to a codon for a specific amino acid in a mRNA. **6** Codons determine the sequence of amino acids in the protein.

Table 1

Template strand	ATG	TCG	TTA	GGG	GGA
Non-template strand	UAC	AGC	AAU	CCC	CCT
mRNA	UAC	AGC	AAU	CCC	CCU
tRNA	AUG	UCG	UUA	GGG	GGA
Amino acid	Tyr	Ser	Leu	Pro	Pro

Table 2

mRNA Codon	tRNA Anticodon	Amino acid	mRNA Codon	tRNA Anticodon	Amino acid
AUU	UAA (ile)	Isoleucine	UGG	ACC	Tryptophan (Trp)
GGA	CCU	Glycine (Gly)	GAA	CUU	Leucine (Leu)
AUG	UAC	Methionine (Met) or START	GAU	CUA	Leucine (Leu)
UCU	AGA	(Serine (Ser))	CCC	GGG	Glycine (Gly)

Conclusion

Protein synthesis consists of two processes: transcription and translation. During transcription, a small section of the DNA double helix unwinds, and the bases on the two strands are exposed. The DNA strand that is transcribed is called the template strand or sense strand. The complementary DNA strand, the one that is not used, is called the non-sense or anti-sense strand. RNA polymerase enzyme binds to the DNA molecule at a region known as the promoter site. This is in front of the gene where transcription will begin. A gene consists of coding regions called exons and non-coding regions called introns. RNA nucleotides (ribonucleotides) line up in the proper order by hydrogen-bonding to their complementary bases on DNA. DNA that includes all the exons and introns of the gene is first transcribed into a complementary RNA copy called 'nuclear RNA', or mature RNA. In a second step, introns are removed from mRNA by a process called RNA splicing. The edited sequence is called 'messenger RNA', or mRNA. This mRNA moves from the nucleus to the cytoplasm and attaches to a ribosome. In the cytoplasm, three-dimensional inverted cloverleaf-shaped molecules called tRNAs are present. At the top of tRNA, a specific amino acid can be attached by means of enzymes called aminoacyl-tRNA synthetases. The tRNAs carry anticodons which are complementary to mRNA codons.

Translation occurs at ribosomes in the cytoplasm. As protein synthesis begins, one small and one large sub-unit of the ribosome come together to form an active ribosome. The first A-U-G codon on the 5' end of the mRNA acts as a 'start' signal for the translation and codes for the introduction of a methionine amino acid. Amino acids continue to be linked until the protein is finished. A special type of bond is formed between amino acids by the enzyme peptidase. The elongation procedure continues until the proper protein is completed. A 'stop' codon (U-A-A, U-G-A or U-A-G) signals the end of the process. An enzyme called the releasing factor then frees the newly made polypeptide chain, also known as the protein from the last tRNA. All mRNA messages are eventually degraded when there is no need to make the protein anymore.



4 Types of inheritance

Part A – Incomplete dominance in snapdragon plants

Results

Table 1 Red × white

W h i t e	Red		
		R	R
	W	RW	RW
	W	RW	RW

Genotypes of F1: All RW
Phenotypes of F1: All pink

Table 2 Pink × pink

p i n k	Red		
		R	W
	R	RR	RW
	W	RW	RW

Genotypes of F1: 1 RR : 2 RW : 1 WW
Phenotypes of F1: 1 Red : 2 Pink : 1 White

Table 3 Red × pink

R e d	Pink		
		R	W
	R	RR	RW
	R	RR	RW

Genotypes of F1: 2 RR : 2 RW
Phenotypes of F1: 2 Red : 2 Pink

Analysis of results

1 Red and white are incompletely dominant. **2 a** Red crossed with white: only pink snapdragons produced. **b** Two pink crossed: red, pink and white snapdragons produced in the ratio of 1:2:1. **c** Red crossed with pink: red and pink snapdragons produced in the ratio of 1:1.

4 Types of inheritance (cont'd)

Part B – Co-dominance in ABO blood groups

Results

Table 4 Homozygous group A × group A

A g r o u p	A group		
		I^A	I^A
	I^A	$I^A I^A$	$I^A I^A$
	I^A	$I^A I^A$	$I^A I^A$

Genotypes: all $I^A I^A$

Phenotypes: all A group

Table 5 Heterozygous group A × group A

A g r o u p	A group		
		I^A	I^O
	I^A	$I^A I^A$	$I^A I^O$
	I^O	$I^A I^O$	$I^O I^O$

Genotypes: 1 $I^A I^A$: 2 $I^A I^O$: 1 $I^O I^O$

Phenotypes: 3 A group : 1 O group

Table 6 Homozygous group A × group B

B g r o u p	A group		
		I^A	I^A
	I^B	$I^A I^B$	$I^A I^B$
	I^B	$I^A I^B$	$I^A I^B$

Genotypes: all AB

Phenotypes: All AB group

Table 7 Homozygous group A × group O

O g r o u p	A group		
		I^A	I^A
	I^O	$I^A I^O$	$I^A I^O$
	I^O	$I^A I^O$	$I^A I^O$

Genotypes: All $I^A I^O$

Phenotypes: All A group

Table 8 Heterozygous group B × group O

O g r o u p	B group		
		I^B	I^O
	I^O	$I^B I^O$	$I^O I^O$
	I^O	$I^B I^O$	$I^O I^O$

Genotypes: 2 $I^B I^O$ and 2 $I^O I^O$

Phenotypes: 2 B group : 2 O group

Table 9 Homozygous group O × group O

O g r o u p	O group		
		I^O	I^O
	I^O	$I^O I^O$	$I^O I^O$
	I^O	$I^O I^O$	$I^O I^O$

Genotypes: All $I^O I^O$

Phenotypes: All O group

Analysis of results

1 Group A and group B alleles are co-dominant. 2 A and B alleles are dominant over the O allele. 3 a 100% have A group b 3 A group : 1 O group
c 100% A group d 2 A group : 2 O group e AB : A : B : O f 100% O group

Part C

Results

Table 10 Colour-blindness: Colour-blind male × normal female

$X^C X^C$	$X^C Y$		
		X^C	Y
	X^C	$X^C X^C$	$X^C Y$
	X^C	$X^C X^C$	$X^C Y$

Genotypes: 2 $X^C X^C$: 2 $X^C Y$

Phenotypes: 2 carrier females : 2 normal males

Table 11 Colourblindness: Normal male × carrier female

$X^C X^c$	$X^C Y$		
		X^C	Y
	X^C	$X^C X^C$	$X^C Y$
	X^c	$X^C X^c$	$X^c Y$

Genotypes: $X^C X^C$: $X^C X^c$: $X^C Y$: $X^c Y$

Phenotypes: 1 normal female : 1 carrier female : 1 normal male : 1 colour-blind male

Table 12 Hemophilia: Hemophiliac male × carrier female

$X^H X^h$	$X^h Y$		
		X^h	Y
	X^H	$X^H X^h$	$X^H Y$
	X^h	$X^h X^h$	$X^h Y$

Genotypes: $X^H X^h$: $X^h X^h$: $X^H Y$: $X^h Y$

Phenotypes: 1 carrier female : 1 hemophiliac female : 1 normal male : 1 hemophiliac male

Table 13 Hemophilia: normal male × carrier female

$X^H X^h$	$X^H Y$		
		X^H	Y
	X^H	$X^H X^H$	$X^H Y$
	X^h	$X^H X^h$	$X^h Y$

Genotypes: $X^H X^H$: $X^H X^h$: $X^H Y$: $X^h Y$

Phenotypes: 1 normal female : 1 carrier female : 1 normal male : 1 hemophiliac male

Table 14 Brown teeth enamel: male with brown teeth enamel × female with white enamel

$X^E X^e$	$X^E Y$		
		X^E	Y
	X^E	$X^E X^E$	$X^E Y$
	X^e	$X^E X^e$	$X^e Y$

Genotypes: $2X^E X^E$: $2X^E Y$

Phenotypes: 2 females with brown teeth : 2 males with white teeth

Table 15 Brown teeth enamel: male with white teeth enamel × heterozygous female with brown teeth enamel

$X^E X^e$	$X^e Y$		
		X^e	Y
	X^E	$X^E X^e$	$X^E Y$
	X^e	$X^e X^e$	$X^e Y$

Genotypes: $X^E X^e$: $X^e X^e$: $X^E Y$: $X^e Y$

Phenotypes: 1 female with brown teeth : 1 female with white teeth : 1 male with brown teeth : 1 male with white teeth

4 Types of inheritance (cont'd)

Analysis of results

1 a Colour blindness: X^C is the dominant allele and X^c is the recessive allele. b Hemophilia: X^H is the dominant allele and X^h is the recessive allele. c Brown tooth enamel: X^E is the dominant allele and X^e is the recessive allele.

Phenotype of parents	Sex-linked recessive inheritance: colour-blindness	Sex-linked dominant inheritance: brown teeth enamel
Affected male × normal female	$X^cY \times X^CX^c$ Girls are carriers, boys are normal	$X^EY \times X^eX^e$ Girls are affected, boys are normal
Normal male × carrier female	$X^CY \times X^CX^c$ 1 normal girl, 1 carrier girl, 1 normal boy, 1 colour-blind boy	$X^eY \times X^EX^e$ 1 girl with brown teeth, 1 girl with white teeth, 1 boy with brown teeth, 1 boy with white teeth
Affected male × carrier female	$X^cY \times X^CX^c$ 1 carrier girl, 1 colour-blind girl, 1 normal boy, 1 colour-blind boy	$X^EY \times X^EX^e$ 2 girls with brown teeth, 1 boy with brown teeth, 1 boy with normal teeth

3 In sex-linked recessive disorders, since males inherit the X chromosome carrying the recessive gene from their mothers and the Y chromosome from their fathers, they are more likely to inherit these disorders from their mothers. 4 In sex-linked dominant disorders, since girls receive an X chromosome from their fathers that has the dominant gene, and one X chromosome from their mothers which is normal, they inherit the disorder.

Conclusion

	Incomplete dominance	Co-dominance	Sex-linked recessive inheritance	Sex-linked dominant inheritance
Location of the gene	autosome	autosome	X chromosome	X chromosome
Phenotype of heterozygote	Alleles blend and an in-between phenotype is expressed	Both alleles are expressed in the phenotype	Girls will be carriers	Girls will be affected
Examples	Red and white snapdragons produce pink colour	A and B blood groups produce AB blood group	Colour blindness Haemophilia	Brown tooth enamel

2 A group can receive blood from A and O groups; B group can receive blood from B and O groups; AB group can receive blood from all 4 groups and is known as a universal receiver; O group can receive blood only from O group and is known as a universal donor.



Pedigree analysis

Part A – Analysis of pedigrees

1 Autosomal dominant inheritance: father has the trait, passed on to both males and females, appears in every generation. 2 Autosomal recessive inheritance: both parents do not show the trait, passed on to both males and females, skipped two generations. 3 Sex-linked recessive inheritance: father has the trait, skipped one generation, more males are affected, more females are carriers and one is affected. 4 Sex-linked dominant inheritance: affected fathers pass on the trait to only females, affected mothers pass on the trait to 50% of male and female offspring. 5 Co-dominant inheritance: A and B produced AB, B, A and O groups because alleles for A and B groups are co-dominant and the allele for O group is recessive to both A and B alleles. 6 Incompletely dominant inheritance: alleles controlling red and white colour are incompletely dominant and produce a third phenotype – roan, two roan parents produced red, roan and white offspring, roan and white parents produced roan and white offspring.

Analysis of results

1 a Genotypes of parents in the 1st generation are HH and hh. b Affected males and females in the 2nd generation: 2 males and 1 female. 3rd generation: 1 male. 2 a Genotypes of the parents in the 1st generation are Cc and Cc. b Affected males and females in the 2nd and 3rd generations: none. 4th generation: 1 male and 1 female. c Cousins. 3 a Genotypes of parents in the 1st generation are X^cY and X^CX^c . b Males are hemizygous because they have only one X and one Y chromosome. c If the mother is a carrier, her sons can inherit the X chromosome carrying the gene for colour blindness. d The father must be colour blind X^cY and the mother must be a carrier X^CX^c to produce a colour blind girl. 4 a Genotypes of the father and the mother in the 1st generation are X^EY and X^eX^e . b All females have brown teeth because they inherit X^E from their father and X^e from their mother. c Since the mother is $X^E X^e$ and the father is $X^e Y$, 50% of their male and female offspring inherit brown teeth. 5 Since both parents are heterozygous IA IO and IB IO it is possible to produce offspring with 4 blood groups: IA IO, IB IO, IA IB, IO IO. 7 The alleles for Red (R) and White (W) are incompletely dominant, so when the offspring inherit RW alleles, they express the in-between colour, roan.

Part B – Making your own family pedigree

Conclusion

A pedigree chart is used to trace the inheritance of a trait over several generations. It shows the relationships among family members and the phenotypes and genotypes of interest for each person in the family. Pedigree charts help people to work out the probability of inheriting a disorder.



Dihybrid inheritance in corn

Hypothesis: In F₁, all the corn kernels will be purple and smooth. In F₂, the ratio of the 4 phenotypes will be 9:3:3:1.

Theoretical results

Table 1 Homozygous dominant (PPSS) × homozygous recessive (ppss)

	PPSS		
		PS	PS
	ps	PpSs	PpSs

Phenotypes: All purple, smooth

Table 2 Heterozygous PpSs × heterozygous PpSs

	PpSs				
		PS	Ps	pS	ps
	PS	PPSS	PPSs	PpSS	PpSs
	Ps	PPSs	PPss	PpSs	Ppss
	pS	PpSS	PpSs	ppSS	ppSs
	ps	PpSs	Ppss	ppSs	ppss

Phenotypes: 9 purple, smooth : 3 purple, wrinkled : 3 yellow, smooth : 1 yellow, wrinkled

Table 3 Test cross: Heterozygous (PpSs) × Homozygous recessive parent

	PpSs			
		PS	Ps	pS
	ps	PpSs	PpSs	PpSs

Phenotypes: 1 purple, smooth : 1 purple, 1 yellow, smooth : 1 yellow, wrinkled

Observed results: Results should be similar to theoretical results.

Analysis of results

1 A dihybrid cross involves the study of inheritance patterns for organisms differing in two traits. 2 Alleles for purple colour and smooth kernels are dominant and alleles for yellow colour and wrinkled kernels are recessive. 3 In F₂ generation, purple, smooth kernels appear most often and yellow, wrinkled kernels appear least often. 4 All the offspring in F₁ generation have purple, smooth kernels. 5 In F₂ generation, the phenotype ratio is 9 purple, smooth : 3 purple wrinkled : 1 yellow smooth : 1 yellow wrinkled. 6 During test crossing, the phenotype ratio is 1 purple, smooth : 1 purple wrinkled : 1 yellow smooth : 1 yellow wrinkled.

Conclusion

1 In a monohybrid cross, only one trait is observed. In a dihybrid cross, the cross between two traits is observed. In a monohybrid cross in F₂ generation the phenotype ratio is 3 dominant : 1 recessive. In a dihybrid cross in F₂ generation the phenotype ratio is 9 dominant dominant : 3 dominant recessive : 2 recessive dominant : 1 recessive recessive. 2 Using a punnet square makes it possible to predict the genotype and phenotype ratios of various crosses.



Patterns of evolution

Part A – Homologous structures due to divergent evolution

Results

Table 1

Bone structure and function	Human (mammal)	Lizard (reptile)	Cat (mammal)	Whale (mammal)	Bat (mammal)	Frog (amphibian)	Bird (aves)
Humerus	Very long	Medium	Long	Thick and short	Thin and small	Long	Medium
Radius	Long	Medium	Long	Thick and short	Thin and small	Thin and short	Thin and long
Ulna	Long	Medium	Long	Thick and short	Thin and small	Thin and short	Thin and long
Metacarpals	Short	Short	Medium	2–3 very long 1–5 short	2–5 very long 1 short	2–5 medium 1 short	4–5 missing 2–3 fused
Habitat	Land	Land	Land	Water	Air	Land, water	Air
Function	To hold and lift	To walk	To walk	To swim	To fly	To hop and swim	To fly

Analysis of results

1 Homologous structures are characteristics that are shared by related species because they have been inherited from a common ancestor. 2 No, they do not have the same function. 3 They have similar bone structure which is modified to suit the function. 4 They evolve due to divergent evolution because they are subjected to different selective pressures.

7 Patterns of evolution (cont'd)

Part B Results

Table 2

Features and function	Dolphin (mammal)	Ichthyosaur (reptile)	Pterodactyl (reptile)	Bird (aves)	Bat (mammal)	Insect (arthropod)
Body shape/ structures present	Stream-lined body Has flippers and tail	Stream-lined body Has flippers and tail	Wing supported by 4th digit Has fibres and hair-like structures	Wing is supported by an elongated radius, ulna and wrist bones and has feathers	Wing is made of a membrane supported by greatly elongated fingers of the hand	Wing is composed of transparent non-living chitin that spans the spaces between a network of wing veins
Habitat	Water	Water	Air	Air	Air	Air
Function	Swimming	Swimming	Flying	Flying	Flying	Flying

Analysis of results

1 All ant eaters have similar anatomical structures due to adaptations to similar eating habits and living in similar environments. 2 No, they do not have a common ancestor. 3 Both live in water and are subjected to the same selective pressures. 4 Convergent evolution gives rise to analogous structures. 5 The wings of birds and insects are analogous because they have same function; flying. 6 When two species have similar anatomical structures that were inherited from the same ancestor, the similarities are referred to as being homologous. When two species have different anatomical structures that were not inherited from the same ancestor but have similar functions, they are referred to as being analogous.

Part C Results

Table 3

Similarities and differences	Placentals	Marsupials
Similarities	Shape and size, habitat, feeding and mode of locomotion	Shape and size, habitat, feeding and mode of locomotion
Differences	Babies develop in uterus and nourished by placenta.	Under developed babies grow in a pouch
Functions	Similar	Similar

Analysis of results

1 Marsupials and placentals resemble in overall shape, mode of locomotion, feeding and foraging because they have adapted to similar way of life and environment. 2 Parallel evolution 3 Geographic isolation and the lack of carnivorous placental mammals allowed for reproductive isolation and random mutation to take over and create a species unique to Australia. 4 Marsupial young are born relatively underdeveloped. In some but not all marsupial species, the mother develops a special pouch on her body in which to nurse her young. In placental mammals, the young develop in the uterus, nourished by a specialised organ attached to the uterus wall – the placenta. 5 a divergent evolution b convergent evolution c parallel evolution

Conclusion

1 Divergent evolution refers to the development of two or more different species from a common ancestor because of environmental adaptations. In convergent evolution, unrelated species become more and more similar in appearance because of a shared environment. Parallel evolution is the development of the same characteristic or adaptation in unrelated organisms due to similar environmental conditions. 2 Homologous structures show that organisms are more closely related because they have similar anatomical features and descend from a common ancestor.



Natural selection in peppered moths

Part A – Experiments on peppered moths

Percentage of moths captured /inference

- 8% dark moths were produced when the larvae were fed with polluted leaves.
- 8% dark moths were produced whether the larvae were fed with polluted leaves or not.
- Moths captured 0%, 100% moths were eaten

8 Natural selection in peppered moths (cont'd)

- Dark moths 6.3%, Light moths 12.5%, Ratio = 1:2. More light moths were recaptured in the unpolluted forest
- Dark moths 27.52%, Light moths 13.14%, Ratio = 2:1. More dark moths were recaptured in the polluted forest.

Analysis of results

1 In Hasebrook's experiment, 8% of the offspring of larvae fed on polluted leaves were dark moths. In experiments conducted in 1930s, 8% dark moths were produced whether the larvae were fed with polluted leaves or not. 2 If the paint is placed over the wings it will come off easily because the moth wings are scaly. 3 Most of the moths could have been eaten by birds or moved out of the area. 4 In the polluted forest, dark moths were better camouflaged against the dark trees and were not eaten by the birds. In the unpolluted forest, dark moths were not well camouflaged against the light coloured trees and so were eaten by the birds. 5 Since there is less pollution and trees have become light coloured again, more dark moths were eaten by the birds and less dark moths were able to breed. 6 Kettlewell concluded that birds acted as selective agents and changed the moth population ratio.

Part B

Results

During pollution

$$\text{If } q = 0.4$$

$$p = 1 - 0.4 = 0.6$$

	$p = 0.6$	$q = 0.4$
$p = 0.6$	0.36	0.24
$q = 0.4$	0.24	0.16

$$(p + q)^2 = (0.6 + 0.4)^2 = 0.36 + 0.48 + 0.16$$

$$\text{Homozygous dark moths} = 0.36 \times 100 = 36$$

$$\text{Heterozygous dark moths} = 0.48 \times 100 = 48$$

$$\text{Total dark moths} = 0.84 \times 100 = 84$$

$$\text{Light moths} = 0.16 \times 100 = 16$$

After pollution

$$\text{If } q = 0.5$$

$$p = 1 - 0.5 = 0.5$$

	$p = 0.5$	$q = 0.5$
$p = 0.5$	0.25	0.25
$q = 0.5$	0.25	0.25

$$(p + q)^2 = (0.5 + 0.5)^2 = 0.25 + 0.50 + 0.25$$

$$\text{Homozygous dark moths} = 0.25 \times 100 = 25$$

$$\text{Heterozygous dark moths} = 0.50 \times 100 = 50$$

$$\text{Total dark moths} = 0.75 \times 100 = 75$$

$$\text{Light moths} = 0.25 \times 100 = 25$$

Analysis of results

1 An allele is the alternative form of a gene. 2 A is the dominant allele and a is the recessive allele. 3 After pollution, the percentage of dark moths decreased from 84% to 75%. The percentage of light moths increased from 16% to 25%. 4 Natural selection. 5 Yes, because dark coloured allele was dominant to light coloured allele.

Conclusion

1 Pollutants did not cause the mutation. The dark phenotype can be produced by the random combination of gametes carrying genes for light and dark colours. 2 During pollution, the frequency of the allele for dark colour increased because dark coloured moths were not eaten by birds. After the pollution cleared, the frequency of the alleles for light colour increased because the white coloured moths were not eaten by birds.



9 Adaptive radiation of finches and tortoises in Galapagos Islands

Part A

Results

Characteristics	Saddle-backed tortoise	Dome-shaped tortoise
Type of shell	Large shell with an opening	Smaller dome shaped shell
Type of neck and legs	Long neck and legs	Very short neck and legs
Type of environment lived	Arid and dry and lower islands	Wet higher islands
Type of food eaten	Tall tree like cacti	Grasses and low lying shrubs

Analysis of results

1 10 million years ago, the ancestors of the giant tortoises arrived on San Cristobal Island by clinging to a piece of driftwood from a river mouth on the Pacific Coast of the South American mainland, and then spread throughout the Galapagos Island. 2 Their number has decreased to 11 sub-species due to excessive hunting and predation by introduced predators. 3 Gigantism evolved due to the lack of predators and the lack of competition for food. 4 Saddle-backed tortoises have a larger body with a high opening in their shell. They have long necks and long limbs. Dome-shaped tortoises have round-shaped shells, very short necks and limbs. 5 The type of environment (dry or wet) and the type of plants they feed on (tall or low lying) acted as selective pressures. 6 Coevolution is the joint change of two or more species in close interaction. Giant tortoises feed on the prickly pear pads of Opuntia. In doing so, they exert a selection pressure for cacti with defence mechanisms to prevent being eaten. This pressure has favoured the tall, tree-form Opuntia with hard bark and spines on islands with tortoises, whereas the shrub form is more common on islands that have never had tortoises.

9 Adaptive radiation of finches and tortoises in Galapagos Islands (cont'd)

Part B

Results

Types of finches	Food eaten	Types of bills
Tree finches	Fruits, insects	Parrot-like bills, grasping bills
Warbler finches	Insects	Probing bills
Ground finches	Seeds	Probing bills, crushing bills

Analysis of results

1 Ancestral finches came from mainland South America. 2 A sub-species is a population which is genetically distinguishable from other such populations and capable of interbreeding with them. If they are of different species they will not be able to interbreed. 3 Thirteen species of finches that exist today differ from each other in body size and in the size and shape of their beaks. 4 Each species has a different shaped beak for feeding on foods such as insects, flowers, leaves, seeds and blood from other animals. 5 Two species of finches can coexist because their feeding habits and habitats could be different. 6 The environment and the type of food acted as selective pressures and gave rise to adaptive radiation.

Conclusion

1 The ancestors of both species arrived on Galapagos Islands from mainland South America. They have evolved into different species over a long period of time due to geographical isolation, reproductive isolation, genetic drift and natural selection. 2 Natural selection gave rise to divergent evolution in these two species.



Comparison of primate features

Results

Table 1 Differences between the skeletons of a chimpanzee and an H. sapien

Features	Chimpanzee	H. sapiens
Backbone: shape	Curved like a C	S-shaped
Rib cage: size and shape	Large and heavy	Smaller and lighter
Skull: size and attachment	Smaller skull attached posteriorly	Larger skull attached below
Arms and legs: length and use	Arms longer than legs and used for walking	Arms shorter than legs and not used for walking

Table 2 Differences between the pelvis, femur, hands and feet of a gorilla, Australopithecus and H. sapien

Features	Ape	Australopithecus	H. sapien
Pelvis: size and shape	Long and narrow	Short and broad	Short and broader
Femur: shape	The femur is shorter and angled out and the ball joint is smaller.	The femur is short and angled in. The ball joint is large.	The femur is longer and angled in. The ball joint is larger.
Fingers in the hand	Curved and longer	Curved and long	Slightly curved and shorter
Opposable thumb	Have opposable thumb	Have opposable thumb	Have opposable thumb
Type of locomotion	Quadrupedal and knuckle walking	Bipedal	Bipedal

10 Comparison of primate features (cont'd)

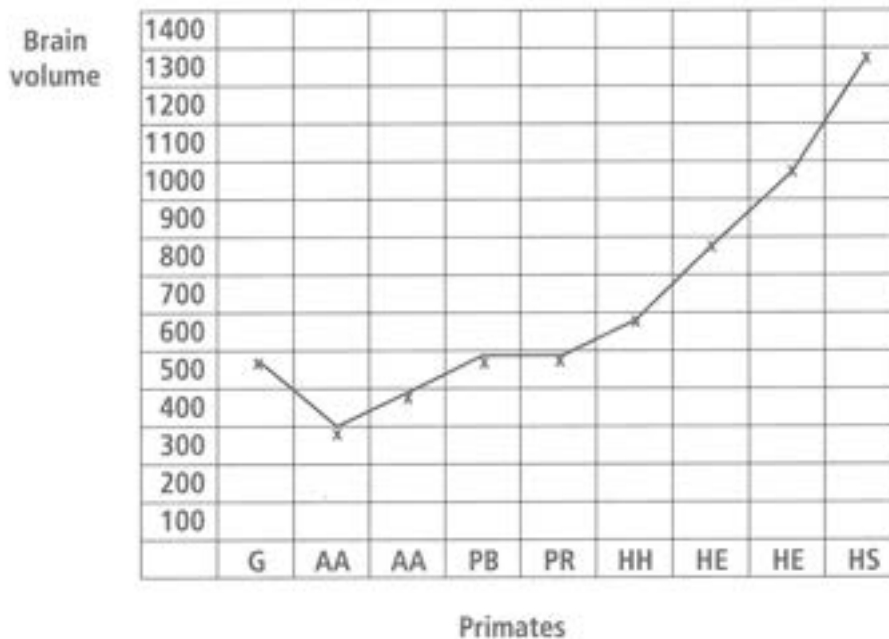
Table 3 Differences in skulls, jaws and teeth of gorilla, Australopithecus and H. sapien

Features	Gorilla	Australopithecus	H. sapien
Skull: shape and size	Narrow and small	Large and medium	Larger and rounder
Brow ridge: size	Large brow ridge	Smaller brow ridge	No brow ridge
Sagittal crest	Present	Absent	Absent
Foramen magnum (the hole below the skull): position	At the back of the skull	Closer to the middle of the skull	In the middle of the skull
Jaw: shape and size	Rectangular and large	In between rectangular and parabolic in shape	Parabolic and small
Forehead and chin: shape and size	Sloping forehead, receding chin	Less sloping forehead and less receding chin	Straight forehead and straight chin
Teeth: type and size	Larger teeth and sharp canines	Medium-sized teeth and canines are not sharp	Smaller teeth and canines are not sharp

Analysis of results

1 S-shaped spine, inferiorly attached skull, bowl-shaped pelvis and inward angled femur. 2 Bipedal locomotion evolved first followed by expansion of brain size. 3 a A. afarensis have the smallest brains and H. sapiens have the largest brains. The size of the brain increased 4-fold. b The brains of Homo species are larger than the brains of Australopithecines.

c



Conclusion

Changes that have occurred during human evolution are: Increase in cranial capacity and brain size; decrease in the size of the jaw and teeth; change in the position of foramen magnum and attachment of the skull to the spine; change in the shape of the spine and ribcage; changes in the shape of the pelvis, femur, and feet to adapt to bipedalism.

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ISBN 978-1-92561-013-0



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